Growth and development:

Which of the following is an etiology of delayed motor development with mental retardation?

a) Benign hypotonia
b) Congenital rubella syndrome
c) Familial and emotional disturbances
d) First degree marasmus
e) Rickets

What is the average weight of a 4-year-old child?

a) 10kg.
b) 12kg.
c) 14kg
d) 16kg
e) 18 kg

At which age the birth height will be doubled?

a) 3 years
b) 4 years
c) 5 years
d) 6 years
e) 7 years

A child was brought to the outpatient clinic for routine checkup of his development. He can stand with support, waves bye-bye but he is not able to walk alone. What is the most likely age of this child?

a) 6 months
b) 8 months
c) 10 months
d) 18 months
e) 24 months

A boy is brought to your clinic to check his developmental progress. He can pedal a tricycle and descends the stairs with alteration of feet. He can also draw a circle but has difficulty with drawing a triangle. What is the most likely age of this boy?

a) 2 years.
b) 3 years.
c) 4 years.
d) 5 years.
e) 6 years

Regarding deciduous (primary) teething: -

a) All 20 primary teeth erupted by the age of 1.5 years
b) Boys teeth earlier than girls
c) Delayed teething to the age of one year is always abnormal
d) Teeth of upper jaw are shed earlier than lower jaw
e) The first deciduous tooth to erupt is central incisor

A child is brought to pediatric clinic for a well-child visit. He can support his head but cannot sit even with support. He smiles on social contact and laughs but cannot take objects to mouth. What is the most likely age of this child?

a) 2 months.
b) 4 months.
c) 6 months.
d) 8 months.
e) 10 months

Which of the following is an average measurement of an infant aged 12 months?

a) 7 kg for weight
b) 11 cm for mid-arm circumference
c) 49 cm for head circumference
d) 55 cm for chest circumference
e) 75 cm for height

The probable age of a developmentally normal child who can sit without support, enjoys mirror, and prefers mother is:

a) 4 months
b) 7 months
c) 9 months
d) 11 months
e) 15 months

A mother brings her first born baby, who is now 7 months old, asking about symptoms associated with teething. You tell her that teething is usually associated with: -

a) Mood disturbances
b) Drooling and sleep disturbance
c) Fever and rash
d) Flushing face or body
e) Diarrhea

A normal child can say mama, dada & respond to sound of name and wave bye-bye. Which of the following motor milestone can be done by this child at this age?

a) Ascend stairs with alternation of feet
b) Climb stairs with one handheld
c) Crawl
d) Descend stairs with alternation of feet
e) Stand on one foot

In the first 4 months of life, the weight increases by: -

a) 100 gm/month
b) 250 gm/month

c) 500 gm/month
d) 750 gm/month
e) 1000 gm/month

The infant doubles his birth weight by:

a) 3 months
b) 5 months
c) 8 months
d) 10 months
e) 12 months

The first ossification center to present at birth is: -

a) Carpal bone
b) Head of humerus
c) Lower end of femur
d) Lower end of humerus
e) Upper end of femur

The first ossification center (Lower end of femur)appears at the age of:

a) At birth
b) 3 months
c) 6 months
d) 1 year
e) 2 years

At the age of 4 years, the ossification centers in the carpal bones equals: -

a) 3
b) 4
c) 5
d) 6
e) 7

A 6-month-old girl comes for checkup with her mother. A look in the girl's mouth reveals no teeth. What is the usual age for eruption of the 1st primary tooth?

a) 2-4 months
b) 4-6 months
c) 6-8 months
d) 8-10 months
e) 10-12 months

As regard normal growth, chest circumference equals head circumference at the age of:

a) 6 months
b) 9 months
c) 12 months
d) 15 months
e) 24 months

Which of the following is the expected weight change for an infant in the first week of life?

a) gain approximately 30 g/day
b) gain approximately 60 g/day
c) lose approximately 5%–10% of birth weight
d) lose of approximately 20-30% of birth weight
e) neither gain nor lose weight

Achild has his deciduous canine erupted, can climb stairs with one handheld and can use 10 – 15 words. His average age is:

a) 9 months
b) 12 months
c) 18 months
d) 24 months
e) 30 months

The mother of a 15-month-ol girl comes to your clinic for her daughter assessment. The girl walks with one handheld, points to her major body parts, and verbalize 10-15 words. This girl: -

a) Has delayed mental but normal motor development
b) Has delayed motor and mental development
c) Has delayed motor but normal mental development
d) Has normal motor and mental development
e) Neither of the above

A mother of a newborn baby asks you about the normal mental development of her baby. You said that: -

a) The infant makes postural adjustment to dressing at 9 months
b) The infant prefers mother at 7 months
c) The infant smiles on social contact at 3 months
d) The infant verbalizes 10-15 words at 1 year
e) The infant waves bye-bye at 6 months

Pubertal changes in girls: -

a) Breast bud appears 6-12 months after pubic hair
b) Growth spurt occurs early in puberty between 11-12 years
c) Menstruation occurs 1 year later to breast bud
d) Occurs later than that of boys
e) The first sign to appear is pubic hair

Pubertal changes in boys are characterized by: -

a) First sign appears before 9 years old
b) Growth spurt occurs early during puberty
c) Occurs 2 years earlier than girls
d) Started by appearance of pubic hair
e) Started by growth of testis > 2.5 cm in longest diameter

Disorders of puberty is considered when: -

a) Absence of secondary sex characters up to 13 years in girls
b) Absence of secondary sex characters up to 12 years in boys
c) Appearance of secondary sex characters at the age of 12years in girls
d) Appearance of secondary sex characters at the age of 13 years in boys
e) Menstruation in 14-year-old girl

Preventive pediatrics

BCG vaccine is characterized by: -

a) Formation of an ulcer at site of BCG indicate infection and anti-tuberculous should be given
b) It gives 100% protection against TB infection
c) It is given in the skin overlying front of thigh
d) It renders tuberculin test positive after 1 week
e) The reaction following BCG vaccine reaches its maximum size at 6-8 weeks

BCG vaccine is characterized by: -

a) Given during the first 3 months of life after negative tuberculin test
b) It is a killed vaccine
c) It is given by subcutaneous injection
d) It is given in the skin overlying the upper 1/3 of left deltoid muscle
e) It gives 100% protection against TB infection

Sabin vaccine is characterized by: -

a) It can be used in immune-compromised individuals
b) It induces systemic and local intestinal immunity
c) It is an inactivated formalin-killed virus vaccine
d) It is monovalent vaccine
e) Not effective with breast feeding

Post-DTaP-vaccination convulsions:

a) Are common in febrile infants and in those with family history of convulsions
b) Doing EEG and CT is mandatory
c) May lead to the development of epilepsy
d) May occur up to 1 week after vaccination
e) When occurs, it is usually severe and prolonged

The adverse reaction which makes us use only DT subsequently and not DPT is:

a) Convulsions
b) Fever

c) Irritability
d) Rash
e) Swelling

The schedule of the DTaP vaccine is characterized by:

a) Booster dose is given at the age of 1 year
b) Never given with other vaccines in the same day.
c) Primary vaccination consists of a single injection at the age of 6 months
d) The first dose is given at the age of 1 month
e) The only DT vaccine is given at the start of each school stage

Measles vaccine is

a) Component vaccine
b) DNA recombinant
c) Killed vaccine
d) Living attenuated vaccine
e) Measles toxoid

Hepatitis B vaccine should be given as

a) Four doses at 0, 2, 4, 6 months of age
b) Single dose
c) Three doses at 2, 4, 6, months of age
d) Three doses, 0, 1, 6 months of age
e) Two doses at least 6 months apart

Which of the following is true regarding hepatitis B vaccine nature?

a) Capsular polysaccharide vaccine
b) Conjugate vaccine
c) Inactivated vaccine
d) Living attenuated vaccine
e) Recombinant DNA vaccine

Newborns of Hbs-Ag positive mothers should receive:

a) The regular vaccination schedule
b) Hepatitis B vaccine immediately at birth and at 1 and 6 months
c) HB vaccine + HB immunoglobulin at birth and the vaccine at 1& 6 months
d) Hepatitis B immune globulin at birth and every month for 6 months
e) Hepatitis B vaccine plus antiviral drug

Indications of hepatitis B vaccine include:

a) Patients with nephrotic syndrome
b) Patients with thalassemia major
c) Home contacts of hemodialysis patients
d) Post-splenectomy

e) History of hepatitis A infection

Which of the following is contraindicated for immunizing an infant who has been diagnosed with Human Immunodeficiency Virus (HIV) infection?

a) DPT vaccine
b) Hepatitis B vaccine
c) Salk polio vaccine
d) Oral polio vaccine
e) Meningococcal vaccine

Regarding vaccination schedule in Egypt:

a) Hepatitis A vaccine is obligatory.
b) Hepatitis B vaccine is given at birth and 2, 4, 6 months
c) Booster DPT dose at 12 months of age.
d) MMR vaccine at 15 months of age
e) Salk vaccine is the standard for poliomyelitis vaccination

Nutrition:

Colostrum is the perfect first food of newborn because:

a) It has high lactose and fat
b) It has higher calories
c) It has low immune globulin
d) It has low protein and vitamin A
e) It is laxative and helps excretion of meconium

A 3-months-old baby is referred with a history of return of small amount of swallowed milk during and sometimes shortly after feeding. History revealed exclusive breast feeding. He has two to three bowel movements per day. His weight is above the 95th percentile and has a normal examination. The most probable diagnosis is:

a) Abdominal colic
b) Gastritis
c) Gastroenteritis
d) Intestinal obstruction
e) Overfeeding

A 2-month-old infant is on breast milk feeding. Unfortunately, his mother developed acute leukemia and is going to have anticancer chemotherapy. You are asked to advice the mother about the future nutrition to the infant.Your recommendation is:

a) Complete cessation of breast milk and start humanized artificial milk
b) Complete cessation of breast milk and start weaning diet
c) Continue breast milk and start humanized artificial milk

d) Continue breast milk and start weaning diet
e) Continue breast milk only till the age of weaning

Indicators of breast milk adequacy include:

a) The baby appears alert and cry after feeding
b) The baby has bowl movement once per day in early weeks
c) The baby increases in weight
d) The baby sleeps less than usual
e) The baby urinates 3 times per day

The MOST objective indicator of adequate breast milk intake by the infant is:

a) Feeding frequency
b) Rate of weight gain
c) State of hydration
d) Stooling pattern
e) Voiding pattern

Which of the following is an indication of artificial feeding?

a) Infant with chronic diarrhea
b) Low birth weight baby
c) Maternal mastitis
d) Maternal engorged breast
e) Maternal psychosis

Which of the following is an indication of artificial feeding?

a) Infants with cystic fibrosis
b) Infants with galactosemia
c) Maternal Hepatitis B
d) Maternal Hepatitis C
e) Maternal mastitis

At a4-month-well child visit,a mother is distressed because her baby's grandmother believes she should have started feeding her baby solid food at 2 months of age.It is generally recommended that infant foods other than milk be introduced into the infant's diet at about what age?

a) 3 weeks
b) 6 weeks
c) 3 months
d) 6 months
e) 1 year

Which of the following statements reflect proper recommendations regarding introduction of complementary (weaning) foods to infants?

a) Avoid introduction of new food while the baby is unwell
b) Introduce legumes first
c) Introduce new type of food every day
d) The new food is given after breast feeding
e) Use bottle to introduce the new food

Breast fed infants are more likely than bottle fed infants to:

a) Achieve higher mental development
b) Develop facial eczema
c) Develop lactose intolerance
d) Develop rickets
e) Show higher bowel infections

The best breast-feeding protocol for infants is:

a) Avoid breast feeding at night
b) Based on the mother' schedule
c) Exclusivelyfor the 1st 2 months of life
d) Feeding every 2 hours
e) Feeding on demand

Which of the following are recommended to encourage successful breastfeeding?

a) Breastfeeding scheduling every 2-3 hours
b) Initiation of breastfeeding within 1 hour of birth
c) Restricting length of breastfeeding time to prevent nipple soreness
d) Use of pacifiers and artificial nipples in between breastfeedings
e) Use of supplemental formula during the early stages of milk production

A10-week-old child weighing 5 kg is being fed commercial infant formula. The mother is concerned she is under feeding her baby. You tell her that, the amount of milk in each feed on average is:

a) 30 ml
b) 60 ml
c) 90 ml
d) 120 ml
e) 150 ml

The daily requirement of vitamin D in infants less than 1 year is:

a) 100 IU
b) 400 IU
c) 600 IU
d) 1000 IU
e) 2000 IU

Nutritional disorders

A three years old child with mid arm circumference of 11cm has:

a) Mild malnutrition
b) Moderate malnutrition
c) Normal body built
d) Overweight
e) Severe malnutrition

A 7-month-old female is presented with poor weight gain. Her weight is 4 kgs. she appears alert with no anorexia but very thin with skin folds on her arms, thighs, and buttocks. Her feeding consists of diluted cow's milk. How many calories will she need in the rehabilitation phase for proper weight gain?

a) 100 Kcal/day
b) 100 Kcal/kg/day
c) 150-220 Kcal/day
d) 150-220 Kcal/kg/day
e) 3000 calories/day

A 2-year-old boy presented with edema of both lower limbs. His history reveals complete weaning on starchy food and sugary fluids since the age of 1.5 years. What would you suspect to find on examination?

a) Emaciated child looking like an old man
b) Erythema, hyper-pigmentation and desquamation of skin
c) Irritable child, cry much and sleeps little
d) Loss of subcutaneous fat from lower limbs
e) Thin wrinkled and inelastic skin

Which of the following is correct to differentiate between marasmus and kwashiorkor?

a) Apathy is present with marasmus
b) Edema is present with marasmus
c) Irritability is present with kwashiorkor
d) Muscle wasting in kwashiorkor is more than in marasmus
e) Subcutaneous fat is preserved with kwashiorkor

A child with severe malnutrition is admitted to the hospital. Which of following is one of theappropriate steps in stabilization phase?

a) Add extra salt in his diet
b) Correct micronutrient deficiencies
c) Give 150-220 Kcal/kg/day
d) Isolate the patient from his parents
e) Parenteral feeding

Actions of vitamin D include the following:

a) It decreases intestinal absorption of phosphorus
b) It decreases renal reabsorption of calcium
c) It has direct effect on mineral metabolism of bone
d) It increases excretion of phosphorus by kidneys
e) It stimulates parathormone secretion

You prescribe vitamin D to a-15-month old girl who is presented with rickets. After 2 – 4 weeks of treatment, the probable clinical sign of healing is:

a) Decrease in the head circumference
b) Disappearance of marfan sign
c) Improvement of appetite and gaining weight

d) Increase in the length by 5 cm
e) The child starts to stand and walk

Which of the following is a biochemical change in active rickets?

a) High serum phosphorus
b) High serum calcium
c) Increased calcinuria
d) Increased phosphaturia
e) Low serum alkaline phosphatase

An 18-month-old infant presents to clinic because of delayed walking. History revealed exclusive breast feeding with little baby food. On examination he has broadening of ends of long bones and palpable swellings at costochondral junctions. What is the expected laboratory finding in this infant?

a) Low serum phosphorus
b) Low alkaline phosphatase
c) Low serum calcium
d) Low serum magnesium
e) Low urinary phosphorus

Early clinical signs of rickets include:

a) Bossing of the head
b) Craniotabes
c) Delayed closure of anterior fontanel
d) Green stick fracture
e) Visible rosary beads

Skull manifestations of active rickets include:

a) Box shaped skull
b) Laterally grown skull
c) Long and narrow skull
d) Small sized skull
e) Upward grown skull

Skull manifestations of active rickets include:

a) Brachycephaly
b) Oxycephaly
c) Caput quadratum
d) Caput succedaneum
e) Scaphocephaly

Hypervitaminosis D is characterized by:

a) Clinical features include polyphagia, vomiting, and diarrhea
b) Features develop 1-3 months after a large dose of vitamin D.
c) Features develop immediately after a large doses of vitamin D.
d) It causes manifestations of increased intracranial tension.
e) Laboratory features include hypocalcemia and hypocalcuria

Treatment of hypervitaminosis D includes:

a) Corticosteroid in severe cases
b) Decreasing the dose of vitamin D
c) Ingestion of large amount of milk
d) Iron chelating agent
e) I V acetylcysteine

Which of the following is a cause of tetany?

a) Acidosis
b) Hypokalemia
c) Hypomagnesemia
d) Hyponatremia
e) Hypophosphatemia

A 15-month-old girl is presented to the emergency department with history of brief generalized convulsions. Upon arrival she was awake, alert and there is no abnormal neurologic finding. The anterior fontanel is about 4 cm, there is 2 upper and lower central incisors, broadening of radius and ulna and positive Marfan sign. The probable cause of the convulsion is:

a) Febrile convulsions
b) Hypocalcemia
c) Hypoglycemia
d) Hypomagnesemia
e) Meningitis

A 23-month-old female is hospitalized for severe malnutrition.Which of the following problems is most likely to be the result of inappropriately rapid intravenous fluid infusion of the severely malnourished child?

a) Congestive heart failure
b) Hyperglycemia
c) Hyperkalemia
d) Hyponatremia
e) Renal failure

Clinical manifestations of severe marasmus include:

a) It is more frequent in older children than in infants
b) There is increased skin elasticity
c) The infant is very sleepy, with little cry
d) There is marked muscle wasting
e) Skin peels off in large scales

A 2-year-old child is presented with weight deficit of 45%, with senile facies and edema of lower limbs. This child has:

a) 1st degree marasmus
b) 2nd degree marasmus
c) 3rd degree marasmus
d) Kwashiorkor
e) Marasmic kwashiorkor

Which of the following is correct about edema in cases of kwashiorkor?

a) It is a pathognomonic manifestation of kwashiorkor
b) It is non pitting edema
c) Ascites is common
d) It is caused by low serum albumin, and altered capillary permeability
e) It is localized to the lower limbs

Edema of kwashiorkor differs from other types of edema in that:

a) Ascites is very rare
a) Associated with hypertension
b) It is caused by heavy proteinuria
c) It starts around eyes
d) May be associated with cyanosis

Which of the following is a Pathognomonic sign of Kwashiorkor?

a) Anemia
b) Anorexia
c) Apathy
d) Edema
e) Hepatomegaly

An 18-month-old child is brought to the clinic. On inspection you note that the child is edematous, has desquamating skin rash over most of the pressure points and has very thin and easily detached hair. What laboratory test is most likely to be present?

a) Heavy proteinuria
b) Hypercholesterolemia
c) Hyperkalemia and hypernatremia
d) Hypoproteinemia
e) Increased serum enzymes as alkaline phosphatase

A 9-month-old female is brought to you because of not gaining weight. On examination, you found that her weight is 5 kg and mid arm circumference is 11.5 cm. She appears alert and hungry. You decide to admit the infant to hospital. How many calories will you initially prescribe for her?

a) 50 kcal/kg/day
b) 100 kcal/kg/day
c) 150 kcal/kg/day
d) 180 kcal/kg/day
e) 220 Kcal/kg/day

Regarding mild malnutrition, which of the following is correct?

a) Associated with increased activity
b) Associated with proper motor development
c) Associated with retardation of mental development
d) It is the commonest form of malnutrition in the 1st6 months of life.
e) It manifests by inadequate physical growth.

Manifestations of acrodermatitisenteropathica include:

a) Arthritis
b) Chronic diarrhea
c) Hepatitis
d) Puffy eyes
e) Rickets

Acrodermatitisenteropathica is due to deficiency of which of the following?

a) Calcium
b) Iron
c) Vitamin A
d) Vitamin D
e) Zinc

Signs of recovery from malnutrition include:

a) Deceleration of linear growth
b) Development of diarrhea
c) Development of hypoglycemia
d) Improvement of appetite
e) Rapid Normalization of weight for age

Treatment of severe malnutrition during the rehabilitation phase includes:

a) Give 100 Kcal/kg/day
b) Stop breast feeding temporarily
c) All patients should be hospitalized till recovery
d) Give parenteral feeding
e) Provide 4-6 gm/kg/day of protein

Treatment of severe malnutrition during the stabilization phase includes:

a) Give 200 Kcal/kg/day
b) Give iron 3 mg/kg/day
c) Give parenteral feeding
d) Give protein 4-6 gm/kg/day
e) Give zinc, copper and multivitamins

Newborn infant

Which of the following item is involved in Apgar score assessment?

a) Blood pressure
b) Heart rate
c) Respiratory rate
d) Temperature
e) Tissue oxygen saturation

Which of the following is true regarding neonatal resuscitation?

a) If at any time the heart rate drops below 100 beats/min, external cardiac compression should be given.

b) If the heart rate drops below 150 beats/min, breathing by mask ventilation should be started
c) Most infants require advanced resuscitation.
d) Providing optimal ventilation is the key to successful neonatal resuscitation.
e) The mouth and nose are vigorously suctioned.

Initial steps of neonatal resuscitation include:

a) Bath the baby.
b) Clear the airway as necessary.
c) Keep the baby neck in fully flexed position.
d) Provide bag and mask ventilation.
e) Provide chest compression.

While examining a newborn, the following is true:

a) Breast enlargement in male newborn denote endocrinal disease.
b) Mongolian spots are blue skin areas that may be observed mainly on the back and are of great pathological significance.
c) Soft systolic murmur always indicates valve lesion.
d) Most of the newborn infants pass urine within 12 hours of birth.
e) Central cyanosis may be seen in normal newborns.

Regarding Moro (startle) reflex, the following is true:

a) Absence of the Moro reflex in a term newborn is meaningless.
b) An asymmetric response may signify a fracture/ dislocation, brachial palsy.
c) Is obtained by placing the prone infant in a knee chest position.
d) It disappears after 2 months.
e) The head is momentarily allowed to fall on the examining table.

Physiologic jaundice is characterized by

a) Hepatomegaly is usually present
b) It can occur at any time of the neonatal period
c) Kernicterus may complicate the condition
d) May persist after the second week of life
e) The newborn is usually of a good general condition

High-risk factors in newborns include:

a) Male sex.
b) Maternal age < 40 years.
c) Maternal age> 16 years.
d) Maternal emotional stresses.
e) Single pregnancy.

General characters of preterm infants include:

a) Lanugo hair is less in preterm than in full term.
b) The chest circumference exceeds that of the head.
c) The nails are short but not necessarily soft.
d) The skin is shiny and has a darker pink color than at term.
e) Weight is more closely related to maturity than is length.

Preterm baby is more prone to hypothermia because:

a) Excess brown fat in preterm infants.
b) Heat production by shivering or muscle activity is optimal.
c) The baby's thermostatic mechanism is poor.
d) The surface area is small in proportion to the body's weight.
e) There is a well-developed cutaneous vascular response.

Common disorders of preterm infants include:

a) Circulatory instability.
b) Hypercalcaemia.
c) Hyperglycemia.
d) Hyperthermia.
e) Meconium aspiration syndrome.

Discharge criteria of preterm babies from the neonatal unit:

a) An average of weight gain of 50-60 gm/day.
b) Current weight of 1200 gm or more.
c) Post conceptual age of about 32 weeks.
d) Stable temperature in an open crib.
e) Taking all nutrition by tube.

Which of the following is a feature of preterm infant?

a) Absence of Lanugo hair
b) Birth weight usually > 2.5 kg
c) Deep creases on the soles of the feet
d) Head circumference is less than the chest circumference
e) Length is more closely related to maturity than is weight

In newborn infant, conjugated hyperbilirubinemia may be caused by:

a) Breast milk jaundice
b) Cephalohematoma
c) Congenital biliary atresia
d) Rh incompatibilities
e) Spherocytosis

Problems encountered in small for gestational age include:

a) Anemia
b) Hyperglycemia.
c) Hyperthermia.
d) Perinatal asphyxia
e) Respiratory distress syndrome

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Which of the following is true regarding tracheoesophageal fistula (TOF)?

a) Affected infants have a lot of mucus with cough and chocking during feeding.

b) Diagnosis is rapidly confirmed by failure to pass a nasogastric tube beyond the anterior naris.
c) It should be suspected in cases with oligohydramnios.
d) The most common type is H shaped type.
e) The presence of air in the abdomen indicates a fistula between the trachea and the proximal esophagus.

Which of the following is true regarding caput succedaneum?

a) Aspiration is needed.
b) It is an area of edema of the soft tissues over the presenting part of the scalp.
c) It usually resolves within several weeks of life.
d) Mostly associated with intracranial hemorrhage.
e) The swelling cannot extend across the middle line.

Which of the following is true regarding cephalohematoma?

a) Aspiration treatment is needed for uncomplicated cephalohematoma.
b) Blood transfusion should be administered in all cases.
c) It can cross suture lines.
d) It resolves within 2 days.
e) Swelling is not visible at birth.

Which of the following is true regarding Erb's paralysis?

a) Elbow is in flexion.
b) Forearm is in supination.
c) Resulting from injury of the seventh and eighth cervical nerves ($C_{7,}$ C_8).
d) The affected arm is held in shoulder abduction and external rotation.
e) Wrist is in flexion

Which of the following match definition of significant neonatal apnea?

a) Any duration if accompanied by tachycardia.
b) Any duration if associated with cyanosis.
c) Longer than 2 seconds.
d) Longer than 5 seconds.
e) Longer than 10 seconds.

Idiopathic apnea of prematurity is characterized by:

a) Central apnea is characterized by persistent chest wall motion.
b) It is common on the 1st day of life.
c) Obstructive apnea is characterized by absent chest wall motion.
d) The incidence of idiopathic apnea of prematurity varies directly with gestational age.
e) The onset of idiopathic apnea occurs on the 2nd – 7th day of life.

Manifestations of central respiratory failure in neonates include:

a) Expiratory grunt
b) Frothing at lips
c) Intercostal retraction
d) Slow, irregular, gasping respiratory efforts
e) Tachypnea

Risk factors for development of respiratory distress syndrome (RDS) include:

a) Infant of diabetic mother.
b) Post term infants.
c) Single pregnancy.
d) Small for gestational age baby.
e) Vaginal delivery.

Respiratory support for neonates with respiratory distress include:

a) Assisted ventilation is indicated in cases with hypoxemia ($PaO_2 \leq 80$mmHg).
b) Assisted ventilation is indicated in cases with $PaCO_2$ 濾60 mmHg and pH<7.2.
c) Oxygen preferably cold and humidified.
d) Surfactant intravenous replacement therapy for cases with hyaline membrane disease
e) The principal goal is to ensure adequate oxygenation but not ventilation.

Causes of neonatal conjugated hyperbilirubinemia include:

a) Cephalhematoma.
b) Congenital biliary atresia
c) Congenital hypothyroidism.
d) Physiologic jaundice.
e) Spherocytosis.

Causes of neonatal unconjugated hyperbilirubinemia include:

a) Congenital biliary atresia.
b) Congenital hypothyroidism.
c) Neonatal hepatitis.
d) Neonatal sepsis.
e) TORCH infections.

Jaundice, that appears within the first 24 hour of life could be due to:

a) ABO incompatibilities.
b) Biliary atresia.
c) Breast milk jaundice.
d) Hypothyroidism.
e) Physiological jaundice.

Clinical manifestations of physiological jaundice include:

a) Disappears at the age of 14-21 days.
b) Hepatosplenomegaly in some cases.
c) Infants are healthy with no abnormal physical signs.
d) Onset at first 24 hours of life.
e) There is marked anemia.

Kernicterus is characterized by which of the following?

a) Asphyxia or sepsis increase the risk of occurrence
b) Bilirubin deposited in the brain cells, particularly the thalamus.
c) Develops when conjugated bilirubin passes the blood brain barrier.
d) Hypertonicityis one of the earliest clinical manifestations
e) Late clinical manifestations include hypotonia.

Hemorrhagic disease of the newborn is characterized by:

a) A normal level of factors II,VII,X,XI.
b) Could be prevented by vitamin K injection at the time of birth.
c) In full-term infants bleeding will be more severe and prolonged than in preterm babies.
d) Occurs in newborn infants at the 1st day after birth.
e) The bleeding time is prolonged.

Neonatal hypoglycemia due to increased metabolic demand could occur in:
a) Infants of mothers with diabetes mellitus.
b) Infants whose mothers use beta blockers.
c) Infants with galactosemia.
d) infants with severe erybroblastosisfetalis.
e) Low-birth weight infants with respiratory distress syndrome.

Clinical signs of neonatal hypoglycemia include:
a) Flushing.
b) Hoarsy cry.
c) Hyperthermia.
d) Jitteriness.
e) Polyuria.

Clinical manifestations of infants of diabetic mother include:
a) Large for gestational age.
b) Dehydrated.
c) Pale facies.
d) Signs of hyperglycemia occur.
e) Signs of Hypercalcemia occur.

Complications of infants of diabetic mother include:
a) Anemia.
b) Genital anomalies.
c) Hypercalcemia.
d) Hyperglycemia.
e) Respiratory distress syndrome.

Metabolic and electrolyte disturbances causing neonatal convulsions include:
a) Hypercalcemia.
b) Hyperglycemia.
c) Hyperkalemia.
d) hypermagnesemia.
e) Hypernatremia.

Clinical features of neonatal hypothermia include:
a) Apathy, refusal of feeding.
b) Hyperglycemia.
c) Pale extremities especially the hands, feet.
d) Polyuria.
e) Tachycardia may occur.

Maternal risk factors for development of neonatal sepsis include:
a) Intrapartum fever more than 37.5°C.

b) Maternal leukocytosis (>8,000/mm^3).
c) Postnatal asphyxia.
d) Premature rupture of membranes (>4 hour).
e) Uterine tenderness.

Neonatal risk factors for development of neonatal sepsis include:

a) Breast fed neonates
b) Early enteral nutrition
c) Large for gestational age
d) Neonates with endotracheal tubes.
e) Post maturity

Clinical features of tetanus neonatorum include:

a) High grade fever.
b) It is an acute hypotonic paralytic illness.
c) Patient is unconscious with hyperexcitability.
d) Spasm of facial and buccal muscles.
e) Suckling and swallowing are unaffected.

A 3–days-old male was presented with bleeding per umbilicus. He was apparently normal but bled excessively when a blood sample was taken. The result revealed normal CBC and C-reactive protein but prolonged prothrombin time. What is the immediate therapeutic step?

a) Intravenous factor IX infusion.
b) Intravenous factor VIII infusion.
c) Intravenous vitamin K infusion.
d) Packed red blood cell transfusion
e) Platelet transfusion.

Which one of the following complications is most likely to be seen in a neonate of diabetic mother?

a) Hyperglycemia
b) Hypermagnesemia
c) Hypokalemia
d) Hyponatremia
e) Polycythemia

Clinical features of tetanus neonatorum include:

a) Acute hypotonic paralytic illness.
b) Fever, jaundice and lethargy.
c) Patient is unconscious with deep respiration.
d) Spasm of facial and buccal muscles.
e) Suckling and swallowing are not affected.

A 3-day-old breast-fed infant presents with yellowish skin. Both the mother and the baby have O +ve blood. The baby's total serum bilirubin is 11.8 mg/dl and direct serum bilirubin is 0.2 mg/dl. The hemoglobin is 17 gm/dl. Platelet count is 278,000/ul. Reticulocyte count is 1.5%. the peripheral smear does not show abnormalities. Which of the following is the most likely diagnosis?

a) Breast milk jaundice
b) Congenital spherocytic anemia
c) Neonatal sepsis
d) Physiologic jaundice
e) Rh or ABO incompatibility

An infant at birth exhibits pink body with blue extremities, pulse rate of 120/min, limp muscle tone, slow irregular respiration and no response to nasal catheter. What is his Apgar score?

a) 2.
b) 4.
c) 6.
d) 8.
e) 10

A 20-day-old boy had circumcision yesterday, but the wound will did not stop bleeding. On examination he is pale and has tachycardia. There is oozing of blood around the circumcision wound. Investigation reveals normal platelet count, normal prothrombin time and prolonged activated partial thromboplastin time. Which of the following is the most probable diagnosis?

a) Acute lymphoblastic leukemia
b) Haemophilia A
c) Hepatic dysfunction
d) Idiopathic thrombocytopenic purpura
e) Vitamin K deficiency

A preterm infant is born via emergency cesarean section. His birth weight is 1075 g. The infant has tachypnea, grunting, intercostal , subcostal retractions, nasal flaring and cyanosis.A chest radiograph reveals diffusereticulo-granular mottling with air bronchogram. What is your diagnosis?

a) Bacterial pneumonia
b) Bronchopulmonary dysplasia
c) Meconium aspiration
d) Pulmonary interstitial emphysema
e) Respiratory distress syndrome

A 4-week-old infant who was born at term without any complications fed well and gained weight for the first 3 weeks after birth. Over the last week, however, his mother reports that he appears hungry but fatigues with feeding and now takes twice as long to complete his feeding as he did 1 week ago. He also breathes fast during his feedings and stops frequently to "catch his breath." Which of the following is the MOST likely explanation for the symptoms in this infant?

a) Aspiration syndrome
b) Congestive heart failure
c) Gastroesophageal reflux disease
d) Inborn error of metabolism
e) Pneumonia

A 1-day- infant who was born by a difficult forceps delivery is alert and active. Shedoes not move her left arm and keeps it internally rotated by her side with

the forearmextended and pronated. Which of the following is an expected clinical finding?

a) Intact both Moro and grasp reflex.
b) Intact Moro and lost grasp reflex
c) Lost both Moro and grasp reflex.
d) Lost Moro and intact grasp reflex
e) Positive Babinski sign

You attended the delivery of an infant weighing 1400 gm is born at 32 weeks.HR:140, RR:80, temp:35° C. the lungs are clear with bilateral breath sounds, there is no murmur. Which of the following is the most important first step in management of this infant?

a) Administer oxygen
b) Do chest x - ray
c) Obtain CBC and differential
d) Place infant under warmer
e) Start tactile stimulation

During the 1-week health supervision visit, a mother who is exclusively breastfeeding asks about vitamin and iron supplementation for her healthy term infant. She explains that her previous child, who was born at 30 weeks' gestation, was discharged with an oral iron supplement and vitamin.Which of the following is the most appropriate oral supplement to initiate for this infant at this visit?

a) Calcium
b) Folic acid
c) Iron
d) Vitamin D
e) Vitamin K

A newborn was examined following a difficult vaginal delivery, his weight is 5 kg, Moro reflex is unilateral. What is the most likely diagnosis?

a) Cephalhematoma
b) Facial palsy
c) Fractured clavicle
d) Intracranial hemorrhage
e) Postmaturity

You attended a normal vaginal delivery of a term baby. The baby was crying and with good tone. Which of the following should be performed first in the delivery room?

a) Clear the airway
b) Dry the baby
c) Provide warmth
d) Stimulate to breath
e) Suction the mouth

How should a neonate with lethargy and a blood sugar of 20mg/dl be treated?

a) 10% dextrose solution
b) 25% dextrose solution
c) 5% dextrose with saline

d) 50% dextrose solution
e) Early breast feeding

Which of the following is the most probable reason for vaginal bleeding in a 3-day-old term female?
a) Child abuse
b) Hemophilia B
c) Transient thrombocytopenia
d) Vaginal foreign body
e) Withdrawal of maternal hormones

A primiparous woman whose blood type is O positive gives birth at term to an infant who has blood group A positive and a hematocrit of 55%. A serum bilirubin level obtained at 36 h of age is 12 mg/dLand direct serum bilirubin is 0.2 mg/dl. Which of the following laboratory findings would be identified in this case?
a) A normal reticulocyte count
b) A positive direct Coombs test
c) Crescent-shaped red blood cells in the blood smear
d) Elevated hemoglobin
e) Low platelet count

An infant is born to a mother who is HBsAg positive. What should be the initial step to prevent transmission of disease to this baby?
a) Check IgM level of HBsAg.
b) Check level of HBsAg.
c) Check the level of HBeAg.
d) Give hepatitis B immune globulin plus hepatitis B vaccine
e) Vaccinate with hepatitis B vaccine.

You attended the delivery of a preterm baby 30 weeks of gestation. Immediately after birth the baby developed respiratory difficulty which increases in intensity thereafter. At 6 hours of age, his RR is 70/min, with grunting, retractions and cyanosis. Which of the following radiological signs is characteristic for this presentation?
a) Air bronchogram
b) Hyperinflation
c) Interstitial infiltrate
d) Lobar opacity
e) Pleural effusion

A term female is born via repeat cesarean section to a 30-year-old woman. Immediately after birth she has mild chest retractions and RR of 70/m. Chest auscultation in the delivery room reveals clear breath sounds. Which of the following is the most appropriate next step?
a) Administer supplemental oxygen and observe
b) Bag-and-mask ventilation.
c) Begin intravenous antibiotic therapy.
d) Deliver surfactant therapy.
e) Endotracheal intubation with direct suction.

Infectious disease:

A 7-year-old child presented with fever and convulsion. Examination showed disturbance in level of conscious, positive Kernig sign, and Brudzinski sign. Which of the following is an expected CSF finding in this case?

a) Clear
b) Glucose is increased
c) Pressure is normal
d) Protein is decreased
e) WBCs increased with neutrophils predominate

Organisms commonly causing meningitis in the first 2 months of age:

a) Group A beta hemolytic streptococci
b) Group Bbeta hemolytic streptococci
c) Neisseria meningitides
d) Staphylococcus aureus
e) Streptococcus pneumonia

A10-week-old infant develops fever (38.5°C), vomiting, and irritability. His heart rate is 170/min and respiratory rate is 40/min. The infant's anterior fontanel is bulged, but there is no neck rigidity. Twelve hours later he developed disturbed level of consciousness and convulsions. What is the most appropriate diagnostic test?

a) CSF examination
b) Complete blood count and ESR
c) Skull X-ray
d) Chest X-ray
e) Serum calcium level

A previously well 10-year-old boy brought to the emergency unit with acute onset of headache and fever. Over the next 2 hours he developed stiff neck, vomiting and altered mental status. On examination: temp >40°C, heart rate 140/min, BP 110/70, an RR 25/min. He is unaware of the surrounding. The most suitable next step to diagnose the case is to order:

a) Complete blood count
b) CSF examination
c) CT brain
d) ESR and C-reactive protein
e) Skull X ray

A previously well 10-year-old boy brought to the emergency unit with acute onset of headache and fever. Over the next 2 hours he developed stiff neck, vomiting and altered mental status. On examination: temp >40°C, heart rate 140/min, BP 110/70, an RR 25/min. He is unaware of the surrounding. What further clinical sign you wish to find on examination?

a) Chvostek sign
b) Ewart sign

c) Flag sign
d) Kernig sign
e) Trousseau sign

Which of the following contacts should receive rifampin chemoprophylaxis after diagnosis of invasive Neisseria meningitidis in a child?

a) All close contacts in the same household or daycare facility regardless of age and immunization history
b) All close contactsin the same household younger than 18 years
c) All close contactsin the same household younger than 4 years
d) All unimmunized or partially immunized close contacts in the same household and in the same day care facility
e) All unimmunized or partially immunized healthcare workers involved in care of the child

A 15-month-old child who attends a child-care center 5 days per week is diagnosed with meningococcal infection. Which of the following agents should be used for prophylaxis for the child-care classmates of this child?

a) Cefixime
b) Chloramphenicol
c) Ciprofloxacin
d) Erythromycin
e) Rifampin

A 3-year-old child presents with fever, diffuse finely papular red rash with circumoral pallor, coated tongue with reddish swollen papillae and follicular tonsillitis of 1-day duration. What is the most likely diagnosis?

a) Avian influenza
b) Measles
c) Roseola infantum
d) Rubella (German measles)
e) Scarlet fever

Rash of scarlet fever is characterized by:

a) Centrifugal maculopapular rash progress to purpura
b) Diffuse punctate or finely papular red rash, blanches on pressure
c) Erythematous, hyperpigmented and desquamated rash
d) Maculopapular rash starts at hairline & behind ears then spread to trunk and limbs
e) Pleomorphic rash consists of macules, vesicles and crusts

You are already examined a 6-year-old boy with fever, vomiting and chills. On examination you find the tonsils hyperemic with gray white exudate. An erythematous rash, blanches on pressure is covering the body except around the mouth. The most probable organism is

a) Group A Streptococci
b) Herpes virus 6
c) Measles virus
d) Rubella virus
e) Varicella zoster virus

The complete blood count in enteric fever is characterized by:

a) Leukocytosis with absolute lymphocytosis
b) Leukocytosis with absolute lymphopenia
c) Leukocytosis with bandemia
d) Leukopenia with relative lymphocytosis
e) Normal leukocytic count with thrombocytosis

Intestinal perforation is a complication of salmonella typhi infection. Which of the following is the most common site?

a) Proximal duodenum
b) Distal ilium
c) Ascending colon
d) Sigmoid colon
e) Rectum and anal canal

A child who has recently returned from a visit to a developing country experiences a 2-week illness characterized by gradually increasing fever with temperature that eventually reaches 39°C, associated with headache, malaise, cough, and abdominal pain. The most likely diagnosis is:

a) Cholera
b) Diphtheria
c) Meningitis
d) Shigellosis
e) Typhoid fever

Clinical presentation of pertussis infection is characterized by:

a) Dry spasmodic cough with expiratory wheeze
b) Gradual onset of cough, followed by abrupt onset of fever and whooping
c) High grade fever followed by croupy cough
d) Mild upper respiratory catarrh followed by paroxysmal cough and whooping
e) Sudden onset of fever and productive cough

Epidemiology of Pertussis is characterized by:

a) Age incidence is high < 6 months due to lack of transplacental protection
b) Age incidence is highest in children above one year of age
c) Majority of cases occur in the winter season
d) Males are affected more than females
e) Pertussis is not a contagious disease

What is the recommended antibiotic for treatment of Bordetella pertussis in a 2-month-old infant?

a) Ampicillin
b) Cefotaxime
c) Erythromycin
d) Gentamycin

e) Vancomycin

After diagnosis of pertussis in a toddler, erythromycin should be given to the patient and to which family members?

a) All family members regardless of age, symptoms, or immunization
b) All non-immunized family members
c) Only those < 7 years old
d) Only those who are symptomatic
e) Only those with compromised immunity

A 2-year-old child is presented with multiple lymph nodes enlargement in the cervical, axillary and inguinal regions. He has history of night sweating, anorexia and weight loss. The lymph nodes are matted and not tender. The possible diagnosis is:

a) Acute lymphoblastic leukemia
b) Infectious mononucleosis
c) Lymphoma
d) Systemic lupus erythematosus
e) TB infection

A 2-year-old child is presented with multiple lymph nodes enlargement in the cervical, axillary and inguinal regions. He has history of night sweeting, anorexia and weight loss. The lymph nodes are matted and not tender. The next step for diagnosis is:

a) Blood culture
b) Blood smear
c) Monospot test
d) Tuberculin test
e) Widal test

False positive tuberculin test may be due to

a) Corticosteroid therapy
b) Congenital immunosuppression
c) Malnutrition
d) Previous vaccination with BCG
e) Very young age

False negative tuberculin test can be found with:

a) Febrile bacterial illness
b) Infection with non-tuberculous mycobacteria
c) Older age group
d) Poor technique of the test
e) Previous BCG vaccination

Which of the following is a true statement concerning the tuberculin skin test?

a) Corticosteroid therapy may increase the amount of reaction
b) It is administered by subcutaneous injection
c) It is nonreactive in up to 50% of infants received BCG
d) Prior BCG vaccination is a contraindication to do tuberculin skin test

e) The reaction is measured 24 after administration

Primary pulmonary TB disease is characterized by

a) Severe productive cough
b) Severe respiratory distress
c) Generalized lymphadenopathy
d) Hepatosplenomegaly
e) More than 50% of patients have no physical findings

In miliary TB disease, which of the following is correct?

a) Chest X ray is not characteristic
b) Generalized lymphadenopathy and hepatosplenomegaly may develop
c) It is most common in well-nourished patients
d) It occurs within 2-6 days of the initial infection
e) Tuberculin skin test is nonreactive in all patients

Indications of steroid therapy in patient with TB include:

a) Miliary TB
b) Perinatal TB
c) Renal TB
d) TB lymphadenitis
e) Tuberculin negative patients

A 5-year-child is exposed to his father who has active TB infection. The proper action for this child is to:

a) Do tuberculin skin test to check infection
b) Give BCG vaccine
c) Give isoniazid chemoprophylaxis
d) Give short course of rifampin and isoniazid
e) Wait and see

Aspirin should not be used as antipyretic in cases of viral infection for the risk of development of:

a) Bleeding tendency
b) Disseminated intravascular coagulation
c) Guillian-Barre syndrome
d) Reye's syndrome
e) Subacutesclerosingpancephalitis

Pleomorphic rash is the feature of:

a) Chicken pox
b) Measles
c) Roseola infantum
d) Rubella
e) Smallpox

The most frequent complications of mumps infection in childhood is:

a) Acute gastroenteritis
b) Acute hepatitis
c) Acute peritonitis

d) Bronchopneumonia
e) Meningoencephalomyelitis

The following is a common complication of Mumps in adolescents:

a) Meningitis
b) Myocarditis
c) Nephritis
d) Orchitis and epididymitis
e) Pancreatitis

Mumps infection is characterized by:

a) Hepatosplenomegaly
b) Generalized lymphadenopathy
c) Painful enlargement of parotid gland
d) Maculopapular rash
e) Mastoiditis

The following are characters of German measles:

a) Rash takes several days to cover the body
b) Koplik spots are pathognomonic for the disease
c) Mild prodromal stage
d) Presence of vesicular rash allover the body
e) Fading of the rash leaves branny desquamation

Roseola infantum differ from measles in that:

a) Hepatosplenomegaly is a characteristic sign
b) Itching is severe
c) Long prodromal stage
d) The rash appears after the fever subsides
e) The rash is of the vesicular type

Congenital rubella syndrome may occur if pregnant mother infection occurs:

a) Before conception.
b) During lactation
c) During the first trimester of pregnancy
d) During the last trimester of pregnancy
e) During the second trimester of pregnancy

Clinical manifestations of congenital rubella syndrome include:

a) Bone deformities
b) Choanal atresia
c) Hydrocephalus
d) Mental retardation and deafness
e) Normal intrauterine growth

As regard to viral influenza infection:

a) Avian flu is associated with mild infection in humans
b) Avian flu is transmitted from infected pigs
c) Classic influenza in young children presents with fever, vomiting and diarrhea

d) Infection with one serotype gives protection from other types
e) Older children are at higher risk of developing swine flu complications

A 3-year-old child is complaining of colicky abdominal pain of 1-week duration. There is diarrhea with tenesmus, blood and mucous in the stool. There is no associated fever. The most probable etiology is:

a) Amoebic dysentery
b) Ascaris infestation
c) Bacillary dysentery
d) Giardiasis
e) Rota virus infection

In intestinal amoebiasis, which of the following is correct?

a) Constitutional symptoms are common
b) Diarrhea is rarely associated with tenesmus
c) High grade fever
d) Stools are bloodstained and contain mucus with few leukocytes.
e) The onset is usually acute

Features favoring diagnosis of shigella dysentery than amoebic dysentery include:

a) Colicky abdominal pain
b) High grade fever
c) Presence of blood and mucus in stool
d) Presence of tenesmus
e) Presence of vomiting

The most morbid feature of giardiasis is:

a) Bleeding tendency
b) Extension to liver
c) Local perforation and peritonitis
d) Malabsorption
e) Upper GI bleeding

Giardiasis should be suspected in patients with:

a) Acute abdomen
b) Hematemesis and melena
c) Intense anal pruritus
d) Malabsorption symptoms
e) Persistent vomiting

Which of the following statements concerning the treatment of Giardia lamblia infection is true?

a) Asymptomatic cyst excreters should be treated
b) Children with malabsorption require treatment
c) Metronidazole is given in a single dose
d) Multiple drug therapy is needed
e) Stool analysis should be repeated to confirm cure

Which of the following is a clinical manifestation of Enterobius Vermicularis worm infestation?

a) Severe anemia
b) Severe abdominal pain
c) Nocturnal anal pruritus
d) Intestinal obstruction
e) Bloody diarrhea

A 6-year-old girl who was previously healthy presents with a 1-week history of nocturnal perianal itching. There are no other symptoms, and findings on physical examination are normal. The most appropriate therapy is:

a) Metronidazole for 7-10 days
b) Tinidazole for 3 days
c) Single oral dose of mebendazole repeated after 2 weeks
d) Iodoquinol for 20 days
e) Neomycin and lactulose for 3 days

Ancylostoma infestation is associated with:

a) B12 deficiency anemia
b) Intestinal obstruction
c) Iron deficiency anemia
d) Liver fibrosis
e) Toxic megacolon

Chronic schistosomamansoni infection may present with:

a) Chronic renal failure
b) Intestinal obstruction
c) Intestinal perforation
d) Portal hypertension
e) Terminal hematuria

Digestive system:

A 10-month-old infant presented with history of diarrhea 15 times / day for two days. You try to exclude metabolic acidosis in this infant. Which of the following criteria of respiration correlate with metabolic acidosis?

a) Alternating shallow and deep breathing
b) Rapid, deep breathing
c) Rapid, shallow breathing
d) Slow, deep breathing
e) Slow, shallow breathing

Aphthus ulcers are characterized by:

a) Common site is the soft palate
b) Extensive vesicles are found around the mouth
c) Lesions are extremely painful

d) Onset usually during early childhood
e) They are of viral origin

A 5-month-old girl who was 7 kgs weight and breastfed, developed frequent diarrheal motions and vomiting for 2 days before her mother brought her to the hospital. On examination, she is 6.5 kgs, irritable, with sunken eyes, dry tongue, and her abdominal skin pinch goes back slowly. She still has repeated vomiting. Your decision will be:

a) Hospitalize her for IV fluid therapy
b) Hospitalize her for nasogastric ORS
c) Hospitalize her for oral ORS
d) Send her home with oral home-made fluids
e) Send her home with oral ORS

Which of the following is a complication of IV fluid therapy for treatment of dehydration?

a) Cerebral thrombosis
b) Gastric dilatation.
c) Paralytic ileus
d) Rapid infusion may lead to heart failure.
e) Repeated vomiting

A 2-year-old child with fever is seen in hospital following a generalized seizure. He also has frequent watery, bloody diarrhea. Which of the following is the most expected causative organism?

a) Giardia Lamblia
b) Entamoeba histolytica
c) Entrobiusvermicularis
d) Shigella
e) Staph aureus

Which of the following signs can be used to detect some dehydration in an infant with severe malnutrition?

a) Anuria
b) Rapid thready pulse
c) Skin pinch goes back very slowly
d) Sunken eye
e) Thirst

A 10-month-old infant presented with history of diarrhea 15 times / day for two days. Examination showed sunken eye, abduction of the hands, flexion of the wrist joint, flexion of the metacarpophalangeal joints, extension of interphalangeal joints and adduction of thumb. Which of the following electrolyte disturbance is expected?

a) Hypernatremia
b) Hypocalcemia
c) Hypokalemia

d) Hyponatremia
e) Hypophosphatemia

A 10-month-old infant presented with history of diarrhea for two days. Which of the following is an indication of intravenous rehydration?

a) Anuria
b) Diarrhea more than 10 times/day
c) Infant refuse taste of ORS
d) Thirst
e) Vomiting 4 times/day

Viral hepatitis A infection in children may be complicated by:

a) Acute fulminating hepatitis
b) Chronic active hepatitis
c) Chronic persistent hepatitis
d) Hepatoma
e) Liver cirrhosis

Mode of transmission of viral hepatitis A infection in children is by:

a) Droplet infection
b) During renal dialysis
c) Feco-oral route
d) Insect bite
e) Through contaminate syringes

IgM anti hepatitis A virus is characterized by:

a) Can cross placenta and gives immunity to the infant
b) Its persistence beyond 6 months defines the chronic infection state
c) Present at the onset of the illness and disappear within 4 months
d) Usually detected 2 months after acute infection
e) Usually persistsfor life after acute infection

Hepatic encephalopathy is caused by:

a) Diminished detoxification of drugs e.g methotrexate
b) Diminished detoxification of herbal bush tea
c) Diminished detoxification of pyrolizidine alkaloids
d) Diminished detoxification of toxic intestinal nitrogenous compounds
e) Diminished detoxification of toxins e.gaflatoxin

A child with hepatic encephalopathy must feed with:

a) High caloric- high protein diet
b) High caloric- low protein diet
c) Low caloric- high protein diet
d) Low caloric- low protein diet
e) Low caloric- no protein diet

One of the causes of pre-hepatic portal hypertension is:

a) Occlusion of hepatic veins
b) Occlusion of inferior vena cava
c) Occlusion of portal vein
d) Occlusion of superior vena cava
e) Veno- occlusive disease

Transudate ascites may be caused by:

a) Nephrotic syndrome
b) Neuroblastoma
c) Peritonitis
d) Polyserositis
e) Tuberculosis

Which of the following is a feature of transudate fluid of ascites?

a) Presence of excess pus cells
b) Presence of few leucocytes
c) Protein > 3gm/dl
d) Specific gravity < 1.015
e) Turbid fluid

Treatment of ascites should include:

a) Low fatdiet
b) Low caloric diet
c) Potassium restricted diet
d) Protein restricted diet
e) Sodium restricted diet

Causes of exudate fluid of ascites include:

a) Abdominal TB
b) Constrictive pericarditis
c) Heart failure
d) Liver cirrhosis
e) Nephrotic syndrome

Functional abdominal pain is characterized by:

a) Abdominal tenderness during examination
b) Associated with vomiting
c) Does not arouse the child from sleep
d) Presence of fever
e) Progressive weight loss

Oral candidiasis is characterized by:

a) Anterior cervical lymph node enlargement
b) Elevated creamy white plaques in the buccal mucosa
c) High grade fever
d) The gingiva is erythematous
e) Yellowish ulcerations on the anterior surface of the tongue

Dysentery may be caused by infection with:

a) Cholera
b) E histolytica
c) Enterovirus
d) Giardia
e) Rota vitus

Acute gastroenteritis with some dehydration is characterized by:

a) Absent tears
b) Moist tongue
c) Skin pinch goes back quickly
d) The child dinks normally
e) The child is lethargic

Acute gastroenteritis with severe dehydration is characterized by:

a) Moist tongue
b) Skin pinch goes back very slowly
c) Tears are present
d) The child drinks eagerly
e) The child is irritable

A 6-month-old breast fed infant has diarrhea since last night. He has no vomiting and his stool is watery, 10 times with no blood. On examination: his skin pinch goes back quickly; his tongue is moist and he drinks normally. The rehydration plan you will follow will be:

a) Intra- venous fluid therapy
b) Naso- gastric rehydration at home
c) Naso- gastric rehydration in the hospital
d) ORS at home
e) ORS in the out-patient clinicof the hospital

An 8-month-old breast fed infant is presented with diarrhea since last night. His stool is watery, 8 times, large amount, with no blood. On examination: His skin pinch goes back slowly; his eyes are little sunken, and drinks very eagerly. The rehydration plan you will follow will be:

a) ORS 20ml/kg in the first 2 hours
b) ORS 50ml/ kg in the first 4 hours
c) ORS 50ml/kg in the first 10 hours
d) ORS 75ml/kg in the first 4 hours
e) ORS 75ml/kg in the first 10 hours

Intravenous rehydration is indicated in a child with acute gastroenteritis and which of the following?

a) Gastric dilatation
b) Irritability
c) Marked thirst
d) Polyurea

e) Refuses the taste of ORS

A child with acute gastroenteritis and signs of shock should receive initially which of the following?

a) IV 10 ml/kgof normal saline over 20 minutes
b) IV 20 m/kg of normal saline over 20 minutes
c) IV 20 ml/kgof normal saline over 40 minutes
d) IV 20 m/kgof normal saline over 60 minutes
e) IV 100 m/kgof normal saline over 20 minutes

Nasogastric rehydration is indicated in:

a) Anuria
b) Persistent vomiting more than 4/ hour
c) Severe abdominal distension
d) Severe anorexia
e) Severe dehydration

About feeding during acute gastroenteritis

a) Breast feedings should be continued
b) No solid food should be administered
c) Rice water and cereals should be prevented
d) Soft drinks must be given
e) Sweetened homemade fluids should be given

A 2-month-old infant came to the gastroenterology clinic with recurrent effortless emesis that dribbles out of the mouth. His weight was 6kg and his clinical examination reveals no detectable abnormalities. The most suitable cause of this condition is

a) Gastroenteritis
b) Gastrointestinal obstruction
c) Immature lower esophageal sphincter
d) Psychogenic
e) Urinary tract infection

A 10-year-old patient comes with a chief complaint of nausea, poor appetite, and fever for one week. She had abdominal pain on her right side below the ribs; and her urine is dark. Examination showed: jaundiced and enlarged tender liver. The rest of her examination is normal. Laboratory data shows raised liver enzymes. The next step for diagnosis is to order:

a) Abdominal ultrasound
b) HAV-IgG
c) HAV-IgM
d) Prothrombin time and concentration
e) Urine analysis

A 5- year-old girl presented with vomiting of fresh and clotted blood. There is no bleeding from other body orifices. On examination: She had pallor but no jaundice, marked abdominal distension, hepatosplenomegaly, ascites, and

obvious abdominal wall veins. The possible cause of hematemesis is:

a) Anaphylactoid purpura
b) Hemophilia
c) Angiomatous malformation
d) Peptic ulcer
e) Portal hypertension

A 5-year-old girl presented with nausea, vomiting, abdominal pain, and dark urine. The mother said that her daughter started to attend a day care center 4 weeks ago. On examination there was jaundice and tender liver. Which of the following tests is the most appropriate to confirm the diagnosis?

a) Measuring liver enzymes.
b) PT& PTT.
c) Liver biopsy.
d) HBsAg.
e) IgM anti-HAV

One-year-old infant brought to ER with severe dehydration and shock. Which of the following therapies would you do first?

a) Blood transfusion.
b) Hydrocortisone IV.
c) Normal saline IV bolus.
d) Plasma transfusion.
e) Start IV antibiotic.

Features suggestive of organic abdominal pain include:

a) Abdominal laxity during examination
b) Absence of other symptoms as fever, diarrhea, dysuria
c) Normal weight gain
d) Pain arouse the child from sleep
e) Pain is usually umbilical or periumbilical

Enlarged spleen is manifested by:

a) It does not move with respiration
b) It is not bimanually palpable
c) It ispulsating on palpation
d) It is resonant on percussion
e) The upper border can be felt

Primary herpetic gingivostomatitis is manifested by:

a) Extensive vesicles are present around mouth
b) Fever and systemic symptoms are absent
c) Gingiva is erythematous and tends to bleed easily
d) No ulcerations are found
e) Regional lymph nodes are not enlarged

Which of the following statements is true about definition of diarrhea?

a) Acute diarrhea has acute onset and lasts less than 14 days
b) Consistency of stool is not a criterion in the definition

c) Excess mucous only with stool is considered dysentery
d) Persistent diarrhea has gradual onset and lasts less than 14 days
e) Well-formed greenish stool is a type of acute diarrhea

Convulsions can complicate diarrhea due to:

a) Brain edema
b) Hypertensive encephalopathy
c) Hypercalcemia
d) Hyperglycemia
e) Isonatremic dehydration

Abdominal distension complicating diarrhea may be due to:

a) Acute renal failure
b) Hepatomegaly
c) Massive ascites
d) Paralytic ileus
e) Urine retention

Use of oral rehydration solution is characterized by:

a) Easily prepared and is used at home
b) Has no role in bloody diarrhea
c) It is given after the 1st year of life
d) It is not effective in hyponatremic dehydration
e) It is not given in the presence of vomiting

Indications of antimicrobial treatment in acute diarrhea include:

a) Diarrhea with persistent vomiting
b) Diarrhea with severe dehydration
c) Lactose intolerance diarrhea
d) Parenteral diarrhea
e) Viral infection diarrhea

Hepatic causes of portal hypertension include:

a) Bilharzial fibrosis
b) Budd- Chiari syndrome
c) Constrictive pericarditis
d) Portal vein thrombosis
e) Tumor compressing portal vein

Pathology of Veno-occlusive disease starts in:

a) Branches of portal vein
b) Hepatic sinusoids
c) Inferior vena cava
d) Large hepatic veins
e) Terminal hepatic venules

A 10-month-old infant presented with history of diarrhea 15 times / day for two days. Examination showed abdominal distention. Which of the following electrolyte disturbance is expected?

a) Hypernatremia
b) Hypocalcemia
c) Hypokalemia
d) Hyponatremia
e) Hypophosphatemia

Respiratory system

Non-pulmonary causes of respiratory distress include:

a) Brain stem disease
b) Hypertension
c) Increased intracranial tension
d) Narcotics
e) Tense ascites

Criteria favoring the diagnosis of hematemesis over hemoptysis include:

a) Alkaline in reaction
b) Bright red blood
c) Coffee ground blood
d) Mixed with air
e) Normal stool

Tactile vocal fremitusis increased in:

a) Atelectasis
b) Bronchial asthma
c) Bronchiolitis
d) Pneumonia
e) Pneumothorax

Which of the following is used in treatment of common cold in children?

a) Broad spectrum antibiotics
b) Intranasal saline drops
c) Large dose of vitamin C
d) Nasal adrenalin inhaler
e) Nasal corticosteroid inhaler

A 2-year-old boy with low grade fever and history of upper respiratory catarrh presented wit respiratory rate of 60/min, intercostal retraction and inspiratory stridor. Which of the following is the most probable diagnosis?

a) Acute non diphtheritic infectious laryngitis
b) Acute epiglottitis
c) Acute laryngotracheobronchitis
d) Acute spasmodic laryngitis
e) Acute bronchiolitis

A 4-year-old boy is admitted with pneumonia. Despite antibiotic therapy, he remains febrile and unwell. Examination showed: respiratory rate 50 breaths/min. The lower right side of chest showed stony dullness on percussion

and reduced breath sounds. Which of the following is the most probable diagnosis?

a) Bronchiolitis
b) Croup
c) Foreign body inhalation
d) Pleural effusion
e) Pneumonia

A 2-year- old girl has upper respiratory catarrh and low grade fever of 2 days. Today she developed cough and noisy respiration. On examination temp: 37.5, respiratory rate: 30/min with prolonged harsh inspiration and clear lung fields on auscultation. She most probably has:

a) Acute bronchiolitis
b) Acute epiglottitis
c) Acute laryngitis
d) Bronchopneumonia
e) Viral pneumonia

A 1-year-old boy is presented with cough, fever, and mild hypoxia. On auscultation, there are crepitations all over chest. Chest X-ray showed pneumatocele. Which of the following is the most likely causative organism?

a) Hemophilus influenzae
b) Mycoplasma pneumoniae
c) Respiratory syncytial virus
d) Staphylococcus aureus
e) Streptococcus pneumoniae

Which of the following criteria can diagnose hemoptysis and exclude hematemesis?

a) Acidic blood
b) Associated with melena
c) Bright red blood
d) Coffee ground blood
e) Food particles with blood

A 4-year-old girl is presented with recurrent cough over the last 2 months. Which of the following could be the etiology of this condition?

a) Bronchiolitis
b) Chronic sinusitis
c) epiglottitis
d) Lobar pneumonia
e) Pleural effusion

Which of the following can cause dullness of Traub's area?

a) Left kidney enlargement
b) Moderate ascites
c) Right hepatic lobe enlargement
d) Right pleural effusion
e) Splenomegaly

A 7-year-old girl presents with difficulty breathing and history of cold for the last 2 days. This is the third time this has happened. Her mother has noticed she 'coughs and whistles' when she goes out in the cold. On examination, she has an upper respiratory infection, respiratory rate 44 breaths/min, mild subcostal and intercostal recession, percussion shows hyper-resonance and scattered wheezes on auscultation. Which of the following is the most probable diagnosis?

a) Bronchial asthma
b) Bronchiolitis
c) Foreign body inhalation
d) Pleural effusion
e) Pneumonia

Which of the following respiratory rhythm match the diseased part of respiratory system?

a) Prolonged expiration in laryngeal disease
b) Prolonged expiration in tracheal disease
c) Prolonged inspiration in bronchiolar disease
d) Prolonged inspiration in laryngeal disease
e) Prolonged inspiration in pulmonary disease

Which of the following viruses is the most commonly associated with viral croup?

a) Adenovirus
b) Human papilloma virus
c) Parainfluenza virus
d) Respiratory syncytial virus
e) Varicella virus

A 4 -year-old child presents to your office with fever, and increased work of breathing manifested by tachypnea and retractions. Chest examination showed decrease chest movement on the left side and central trachea. The upper part of the left side of chest showed: increased tactile vocal fremitus, impaired percussion node, decreased air entry and bronchial breathing. Which of the following is the most likely etiology in this child?

a) Chlamydophila pneumonia
b) Mycoplasma pneumonia
c) Respiratory syncytial virus
d) Staphylococcus aureus
e) Streptococcus pneumonia

A 9-month-old boy presents with fever and difficulty breathing for the last 3 days. His breathing is now interfering with his feeding. He has not had any previous illnesses and his growth is normal. Examination of his chest: respiratory rate 70 breaths/min, hyperinflated chest, subcostal and intercostal recession and scattered wheezes throughout both lung fields. Which of the following is the most probable diagnosis?

a) Bronchiolitis
b) Croup
c) Foreign body inhalation
d) Pleural effusion
e) Pneumonia

A 3-year-old boy, presents with fever and difficulty breathing, getting worse for the last 3 days. Examination shows: Respiratory rate 40 breaths/min. The upper right chest examination showed impaired note on percussion, bronchial breathing and inspiratory crepitations. Which of the following is the most probable diagnosis?

a) Bronchiolitis
b) Bronchopneumonia
c) Foreign body inhalation
d) Pleural effusion
e) Lobar pneumonia

A 3-year-old boy was eating 2 days ago when his younger brother pushed him over. Hewas coughing all over the day. Today, he has been coughing and becomes breathless. He is afebrile. Examination showed: respiratory rate 36 breaths/min, heart apex beat displaced to right and reduced air entry on the right chest. Which of the following is the most probable diagnosis?

a) Bronchiolitis
b) Croup
c) Foreign body inhalation
d) Pleural effusion
e) Pneumonia

A 4-year-old child complains of ear pain. He has a temperature of 102.1°F (38.9°C) and has had a cold for several days, but he has been eating well and his activity has been essentially normal. What is the most common causative organism for this illness?

a) Bordetella pertussis
b) Parainfluenza virus
c) Respiratory syncytial virus
d) Rota virus
e) Streptococcus pneumonia

A 10-month-old boy presents to the Emergency Department with a 2-day history of fever and runny nose. He has been otherwise well. During the night he gradually developed a barking cough in association with a loud noise on inspiration. On examination he has a temperature of 38° C and noisy inspiration accompanied by marked sternal recession. Which of the following is the next step in management?

a) Assess serum calcium level
b) Hospitalization for laryngoscopy in the operating room
c) Hospitalization for O2, IV fluids and adrenaline inhalation
d) Order chest X ray

e) Send him home with antibiotic prescription

A 4-month-old infant, has rapid, labored breathing that has been getting worse over the last 2 days with runny nose. His mother is concerned as she is struggling to get him to feed. On examination he has a temperature of 37.4° C and a respiratory rate of 60 breaths/min. He is coughing. His chest is hyperinflated with marked intercostal recession. On auscultation there are generalized wheezes. Which of the following is the appropriate next course of action?

a) A short course of oral steroids
b) Chest tube placement
c) Emergent intubation and antibiotics
d) Inhaled epinephrine and a single dose of steroids
e) Monitoring oxygenation and fluid status

A 4-month-old infant, has rapid, labored breathing that has been getting worse over the last 2 days with runny nose. His mother is concerned as she is struggling to get him to feed. On examination he has a temperature of 37.4° C and a respiratory rate of 60 breaths/min. He is coughing with marked intercostal recession. On auscultation there are generalized wheezes. Which of the following roentgenographic findings is the most probable expected finding on chest X-ray?

a) Hilar adenopathy
b) Hyperinflation
c) Lower lobe infiltrates
d) Multilobar infiltrate
e) Pleural effusion

A 4-month-old infant, has rapid, labored breathing that has been getting worse over the last 2 days with runny nose. His mother is concerned as she is struggling to get him to feed. On examination he has a temperature of 37.4° C and a respiratory rate of 60 breaths/min. He is coughing. His chest is hyperinflated with marked intercostal recession. On auscultation there are generalized wheezes. Which of the following is the most probable causative organism?

a) Hemophilus influenza
b) Parainfluenza virus
c) Respiratory syncytial virus
d) Staphylococcus aureus
e) Streptococcus pneumonia

A 5-month-old infant is admitted to hospital with difficulty breathing and poor feeding. On examination he has temperature of 38.9° C, a respiratory rate of 50 breaths/min with widespread crackles on auscultation of the chest. Which of the following is the most probable diagnosis?

a) Bronchiolitis
b) Croup
c) Empyema
d) Pneumonia
e) Pneumothorax

Viral pneumonia differs from bacterial pneumonia in that:

a) The associated fever is > 39° c
b) The cough is usually productive
c) The C-reactive protein is increased
d) The pathologic type of pneumonia is usually lobar
e) The WBC count is < 15 000 mostly lymphocytes

The following is an indication of hospitalization of patients with pneumonia:

a) All cases of lobar pneumonia
b) Pneumonia in patients older than 6 years
c) Pneumonia with grunting
d) Pneumonia with sonorous rhonchi
e) Viral etiology of pneumonia

Which of the following is a feature of foreign body inhalation?

a) Common in children older than 5 years.
b) Most of aspirations occur in left lung.
c) Most patients have a negative history.
d) Partial obstruction may cause atelectasis
e) There is immediate episode of chocking.

A 3-year-old boy was playing with small objects when he suddenly starts chocking, gagging with paroxysmal cough. The immediate next step for this case is:

a) A dose of bronchodilator
b) Abdominal thrust
c) Back blows and chest thrust
d) Chest X ray
e) Drinking water

Diagnosis of childhood asthma includes the following EXCEPT:

a) A positive family history of allergy may be present
b) An attack can be induced by exercise.
c) History of eczema
d) History of improvement with asthma medication
e) Symptoms are worse at daytime

Laboratory findings in cases of bronchial asthma include:

a) Eosinophilia
b) Leukocytosis
c) Lymphopenia
d) Raised ESR
e) Thrombocytosis

Quick relieve medications of bronchial asthma include:

a) Inhaled anticholinergic
b) Leukotriene modifiers
c) Long acting inhaled B agonist
d) Nonsteroidal anti-inflammatory drugs
e) Sustained release theophylline

Purulent pleurisy (Empyema) is treated with:

a) Bronchoscopic aspiration
b) Chest physiotherapy
c) Closed drainage under water seal
d) Open thoracotomy
e) Repeated thoracentesis

Foreign body inhalation may cause all of the following EXCEPT:

a) Atelectasis
b) Emphysema
c) Empyema
d) Pneumonia
e) Stridor

Which of the following clinical manifestations favor the diagnosis of lobar pneumonia over bronchopneumonia:

a) Bronchial breath
b) Fever
c) Normal TVF
d) Respiratory distress
e) Shifted mediastinum

Which of the following clinical manifestations favor the diagnosis ofbronchopneumonia over lobar pneumonia:

a) Bronchial breath
b) Age < 2 years
c) Increased TVF
d) Respiratory distress
e) Shifted mediastinum

Which of the following manifestations favor the diagnosis of viral pneumoniaover bacterial pneumonia:

a) Dry spasmodic cough
b) Increased C-reactive protein
c) Leukocytic count > 15000
d) Marked toxicity
e) Temperature > 39C

Central mediastinum is found in which of the following chest diseases?

a) Empyema
b) Lobar pneumonia
c) Pleuropulmonary fibrosis
d) Pneumothorax
e) Total lung collapse

Bronchial breathing is heard in:

a) Bronchial asthma
b) Bronchiolitis
c) Emphesyma

d) Pneumothorax
e) Upper border of effusion

Which of the following signs denoting hearing of bronchial breath?

a) Babiniski sign
b) Chvostek sign
c) Corrigan sign
d) Ewart sign
e) Marfan sign

Which of the following clinical manifestations favor the diagnosis of bronchiolitis over bronchopneumonia:

a) Bronchial breath
b) Dullness on percussion
c) Fever
d) Hyperinflation
e) Respiratory distress

Bronchial breathing is heard in:

a) On the left lung base posteriorly in pericardial effusion
b) Bronchopneumonia in young children
c) Foreign body inhalation with check valve mechanism
d) Normally on the back below thoracic 4 spine
e) Over an emphysematous lung lobe

Cardiovascular system:

A plain X-ray was done to a child and you notice double contour of the right border of the heart. Which of the following is the significance of thisobservation?

a) Biventricular enlargement
b) Left atrial enlargement
c) Left ventricular enlargement
d) Right atrial enlargement
e) Right ventricular enlargement

Which of the following murmurs match the valvular lesion?

a) Continuous murmur in mitral stenosis
b) Early diastolic murmur in pulmonary stenosis
c) Ejection systolic murmur in aortic stenosis
d) Mid-diastolic murmur in aortic regurge
e) Pansystolic murmur in mitral stenosis

Which of the following lesions gives a Mid Diastolic Murmur?

a) Aortic regurgitation.
b) Aortic stenosis

c) Mitral stenosis.
d) Pulmonary stenosis
e) Ventricular septal defect.

Which of the following is a character of the jugular venous pressure?

a) Anterior to sternomastoid
b) Decrease with pressure on liver.
c) It is assessed while the patient lie down
d) More visible than palpable.
e) Undefined upper level.

Isolated right ventricular hypertrophy is found with:

a) Double mitral lesions
b) Isolated aortic regurge
c) Isolated aortic stenosis
d) Isolated mitral regurge
e) Isolated mitral stenosis

A 7-year-old girl presents with a tender and swollen right knee as well as a more recently appearing swollen left wrist. She also has a fever. ECG showed prolonged P-R interval and ASOT is positive. This patient fulfills which of the following modified Jones criteria?

a) 1 Major & 2 minor criteria
b) 1 Major & 3 minor criteria
c) 2 major & 2 minor criteria
d) 2 major & 1 minor criteria
e) 3 minor criteria

An ejection systolic murmur, heard most intensely over the second intercostal space to the left of the sternum followed by accentuated second heart sound. This is most suggestive of which of the following?

a) Aortic stenosis.
b) Atrial septal defect
c) Pulmonary hypertension.
d) Patent ductus arteriosus
e) Ventricular septal defect

Which of the following signs is positive in rheumatic chorea?

a) Hyperreflexia of knee jerk
b) Hypotonia of the muscles
c) Persistent protruding tongue
d) Spastic gait
e) Spooning of nails on extension of fingers

In which of the following cardiac lesions you can auscultate continuous murmur?

a) Aortic stenosis
b) Atrial septal defect
c) Patent ductus arteriosus

d) Tetralogy of Fallot
e) Ventricular septal defect

A 6-week-old infant had tachycardia, tachypnea, and poor feeding. Physical examination revealed BP 80/20 mmHg, a thrill at the 2nd left intercostals space and a continuous murmur. What is the most likely diagnosis?

a) Aortic stenosis
b) Atrial septal defect
c) Patent ductus arteriosus
d) Pulmonic stenosis
e) Ventricular septal defect

A normal 6-month-old infant has a continuous cardiac murmur over the 2nd left space radiating to the left clavicle and bounding peripheral pulses. The proper management step should be taken for this case is:

a) Karyotype evaluation
b) Reassuring of the parents
c) Repeating examination at the age of 12months
d) Search for other congenital anomalies
e) Surgical or catheter correction of the defect

Which of the following is the classic chest x-ray finding in patients with tetralogy of Fallot?

a) Cardiomegaly and decreased pulmonary vascular markings
b) Cardiomegaly with apex cutting the diaphragm
c) Double contour of right border of heart
d) Prominent pulmonary artery and increased pulmonary vascular markings
e) Upturned ventricular apex and decreased pulmonary vascular markings

Which of the following would be expected in a 6-month-old child with big ventricular septal defect?

a) A continuous cardiac murmur
b) Blue clubbing
c) Growth failure
d) Permanent cyanosis
e) Wide pulse pressure

A 2 year old boy is presented with fever and cough of 2 days duration. There was a similar condition 2 month ago. On examination there are signs of respiratory distress, bilateral scattered crepitations over the chest and a harsh pansystolic murmur over the left parasternal border. The diagnosis is:

a) Bronchopneumonia complicated by heart failure
b) Infective endocarditis
c) Large patent ductus arteriosus
d) Rheumatic carditis with left sided heart failure
e) VSD complicated by bronchopneumonia

Which of the following is a component of Tetralogy of Fallot?

a) Biventricular hypertrophy
b) Left ventricular dilatation

c) Left ventricular hypertrophy
d) Right ventricular dilatation
e) Right ventricular hypertrophy

The following manifestation occurs in Tetralogy of Fallot:

a) Bulging precordium
b) Cyanosis which is present at birth
c) Heart failure usually present
d) Paroxysmal hypercyanotic spells
e) Recurrent chest infection

An 8-year-old boy diagnosed as having polyarthritis and carditis. ECG examination revealed presence of prolonged P-R interval. This patient fulfills which of the following modified Jones criteria?

a) 1 Major and 1 minor
b) 1 Major and 2 minors
c) 2 Major and 1 minor
d) 2 Majors
e) 2 Minors

By which of the following guidelines you can diagnose recurrent rheumatic fever?

a) 1 major + 1 minor + evidence of recent streptococcal infection
b) 1 major criteria + evidence of recent streptococcal infection
c) 1 minor criteria + evidence of recent streptococcal infection
d) 2 minor criteria + evidence of recent streptococcal infection
e) 3 minor criteria + evidence of recent streptococcal infection

A 12-year-old boy has migratory arthritis with red, warm, and swollen joints. He has serologic evidence of recent group A streptococcal infection. Arthritis in this condition is characterized by which of the following?

a. Appears after the fever subsides
b. Heals without deformity
c. Involves large and small joints equally
d. Non-painful
e. Seen only in patients with concurrent carditis

A 10-year-old child presented with fever and fleeting arthritis. Examination showed: weak pulse, pulsus paradoxus, congested non pulsating neck veins, apex beat not palpable and heart sounds are distant. These signs are diagnostic of which of the following?

a) Endocarditis
b) Left ventricular heart failure
c) Myocarditis
d) Pericardial effusion
e) Right ventricular heart failure

The auscultatory features of rheumatic carditis include:

a) Continuous cardiac murmur

b) Ejection systolic murmur over the 2nd left space
c) Ejection systolic murmur over the 2nd right space
d) Loud first heart sound
e) Mid diastolic murmur over the apex

What is expected on chest auscultation in a patient with mild mitral insufficiency?

a) High pitched decrescendo early diastolic murmur over 2nd left space
b) High-pitched holosystolic murmur at the apex that radiates to the axilla
c) Loud 1st heart sound and long diastolic murmur over the apex
d) Soft, blowing systolic ejection murmur at the upper left sternal border
e) Soft, blowing systolic ejection murmur at the upper right sternal border

Clinical manifestations of valvular aortic stenosis include:

a) Big pulse volume
b) Corrigan's sign
c) Pan systolic murmur
d) Right ventricular hypertrophy
e) Sustained apex

Clinical manifestations of pericardial effusion include:

a) Accentuated heart sounds
b) Changeable murmurs
c) Congested pulsating neck veins
d) Hyperdynamic precordium
e) Pulsus paradoxus

Which of the following is a feature of pure rheumatic chorea?

a) ESR and ASOT are elevated
b) It is associated with hypertonia
c) It is one of the minor criteria of rheumatic fever
d) It is sufficient alone to diagnose rheumatic fever
e) It results in permanent neurological sequel

A 10-year-old girl presented to outpatient clinic with semi-purposeful, rapid and irregular movements affecting both upper limbs. The mother noticed that her girl cry much without obvious cause and her hand writing deteriorated 2 weeks ago. You suspect rheumatic chorea. What other clinical sign you search for to support diagnosis?

a) Dinner fork deformity of extended hands
b) Hyperreflexia of knee jerk
c) Pale clubbing of finger nails
d) Presence of subcutaneous nodules
e) Signs of acute arthritis

Which of the following is the evidence of streptococcal infection for diagnosis of rheumatic fever?

a) Elevated ESR
b) Elevated levels of interleukins

c) Increasing streptococcal antibody titer
d) Presence of leukocytosis
e) Recent elevation of CRP

Which of the following is a character of rheumatic polyarthritis?

a) It involves mainly small joints
b) It migrates from one joint to another
c) It progresses to residual chronic arthritis
d) It should be treated with prednisone
e) There is expected partial and slow response to salicylates

A 6-year-old girl complains of joint pain in her elbows and wrists. She has had fever for the past 4 days. Past history reveals a sore throat 2 weeks ago. Physical examination is remarkable for swollen, painful joints and a heart murmur but no heart failure. CXR looks normal. Laboratory tests show elevated ASOT. Which of the following drugs is recommended for initial therapy?

a) Digoxin
b) Furosemide
c) Paracetamol
d) Prednisolone
e) Salicylates

Skin manifestations of infective endocarditis include:

a) Circumoral pallor
b) Cold mottled skin
c) Palmar erythema
d) Spider angioma
e) Splinter hemorrhage

A 6-year-old girl – a known case of repaired congenital heart disease - presented after dental procedure, with a complaint of fever, arthralgia, weakness, petechiae, and splenomegaly. After doing your workup you decided to start on antibiotic regimen. Which of the following is the right principle of antibiotic regimen for this patient?

a) Bacteriostatic rather than bactericidal agents are used
b) No need to use empiric antibiotics
c) Oral route of administration
d) Short course 5-10 days
e) Synergistic combination rather than single antibiotic

Which of the following is the recommended antibiotic for primary prevention of rheumatic fever?

a) Benzathin penicillin IM once
b) Cefaclor orally twice daily for 10 days
c) Cefotaxime twice daily for 10 days
d) Sulfadiazine orally once daily for 5 days
e) Tetracyclin once daily for 7 days

A12-year-old child with a surgically corrected pulmonary shunt is to undergo an invasive dental procedure .Which of the following antibiotics would be most appropriate for prophylaxis against bacterial endocarditis?

a) Amoxicillin
b) Cephalosporin
c) Erythromycin
d) Gentamycin
e) Vancomycin

In tetralogy of Fallot the murmur is

a) Early diastolic murmur at mid-to upper left sternal border
b) Ejection systolic murmur at mid-to upper left sternal border
c) Late systolic murmur at upper left sternal border
d) Mid-diastolic murmur at the apex
e) Pansystolic murmur at mid-to upper left sternal border

Asymptomatic 18-month-old child with accidentally discovered loud, harsh left parasternal pansystolic murmur, heard at 3rd and 4th intercostal spaces accompanied by a thrill, so he has:

a) Innocent murmur
b) Large VSD
c) Small VSD
d) PDA
e) Tetralogy of Fallot

An infant with prominence of left precordium and cardiomegaly, a systolic thrill, left parasternal pansystolic murmur at 3rd and 4th intercostal spaces and an apical rumbling mid-diastolic murmur, so he has:

a) Large PDA
b) Large VSD
c) Small PDA
d) Small VSD
e) TGA

In tetralogy of Fallot, cyanosis usually appears within:

a) The first few days of life
b) The first few months of life
c) The first few weeks of life
d) The second year of life
e) The third year of life

A known child with congenital heart disease with shunt developed dyspnea with effort increases with age, cyanosis, small pulse volume, right ventricular lift, pansystolic murmur of tricuspid regurge, and loud second sound. So he developed:

a) Cor pulmonale
b) Eisenmenger syndrome
c) Heart failure
d) Infective endocarditis

e) Pulmonary hypertension

The etiology of Eisenmenger syndrome is characterized by:

a) It complicates rheumatic carditis
b) It is a congenital cyanotic heart disease
c) It is acquired cyanotic heart disease
d) It occurs on top of infective endocarditis
e) No preceding pulmonary hypertension

The mother of a 10-month-old infant tells you that her son becomes blue when he is crying and the frequency is increasing. During examination you palpates a thrill at the left lower sternal border and auscultates a loud systolic murmur in the same area. What would be the most likely cause of these findings?

a) Atrial septal defect.
b) Patent ductus arteriosus.
c) Tetralogy of Fallot.
d) Transposition of great arteries
e) Ventricular septal defect.

By looking to the PA viewchestX-ray of a 5-years-old child, you note an increase in the cardiac size, straight left cardiac border, and the cardiac apex shifted downward and outward with obtuse left cardio-phrenic angle. You interpret these results as:

a) Dilated left atrium and left ventricle.
b) Dilated left atrium.
c) Dilated right atrium and right ventricle
d) Dilated right atrium.
e) Dilated right ventricle.

A 6-hour-old boy presented with marked cyanosis and tachypnea. Chest X-ray shows cardiomegaly and plethoric lungs. Which one of the following is the most diagnosis?

a) Atrial septal defect.
b) Patent ductus arteriosus.
c) Tetralogy of Fallot.
d) Transposition of great arteries
e) Ventricular septal defect

A 5-week-old boy presented to the ER with breathlessness and poor feeding for the last week. O/E patient is ill, temp.37°C, pulse 140/min., RR 65/min. There is active precordium, chest recessions and pansystolic murmur on the lower left sternal border. Which of the following is the best initial treatment?

a) Adrenaline.
b) Captopril.
c) Digoxin.
d) Frusemide.
e) Propranolol.

Diseases of the blood and malignancies

Which of the following is presented with normocytic, normochromic anemia?

a) Folic acid deficiency
b) Iron deficiency
c) Lead poisoning
d) Spherocytosis
e) Thalassemia

Which of the following is presented with microcytic, hypochromic anemia?

a) Acute hemorrhage
b) Folic acid deficiency
c) G6PD deficiency
d) Spherocytosis
e) Thalassemia

Increased levels of fetal Hb can be found in:

a) Acute leukemia
b) G6PD deficiency
c) Iron deficiency anemia
d) Spherocytosis
e) Thalassemia

Which of the following is the most common etiology of acquired pancytopenia?

a) Bone marrow replacement
b) Epstein Barr virus infection
c) Exposure to drugs
d) Exposure to irradiation
e) Idiopathic type

Complete blood count (CBC) in cases of pancytopenia shows:

a) Leukocytosis
b) Reticulocytopenia
c) Reticulocytosis
d) Thrombocytosis
e) Within normal hemoglobin

The only curative therapy for resistant aplastic anemia is:

a) Anti-thymocytes
b) Bone marrow transplantation
c) Cyclosporine
d) Hematopoietic colony stimulating factor
e) Proper antibiotics

The most common type of anemia in our country is:

a) B12 deficiency anemia
b) Folic acid deficiency anemia

c) G6PD deficiency anemia
d) Iron deficiency anemia
e) Thalassemia

Pica is a manifestation of:

a) Acute leukemia
b) Folic acid deficiency anemia
c) Iron deficiency anemia
d) Pancytopenia
e) Thalassemia major

Chronic blood loss causing iron deficiency anemia include:

a) Acute gastroenteritis
b) Ankylostoma infestation
c) Appendicitis
d) Enterobius infestation
e) Giardiasis

You see a toddler with a mild anemia. His diet consists mainly of cow's milk, fruit, and hot chips. He has no supplementation. Which of the following is the probable cause of his anemia?

a) B12 deficiency
b) Folic acid deficiency
c) Iron deficiency
d) Vitamin C deficiency
e) Vitamin D deficiency

Complete blood count (CBC) in iron deficiency anemia shows:

a) Increased mean corpuscular volume (MCV)
b) Increased red cell distribution width (RDW)
c) Increases RBC number
d) Normal mean corpuscular hemoglobin (MCH)
e) Thrombocytopenia

Which of the following is correct as regard iron therapy?

a) A course of 10-15 days is enough in most patients
b) Dose is 60 mg/kg/day of elemental iron
c) Improved appetite is a late response within 2-3 weeks
d) It is given by IM injection
e) It is given orally between meals

Anemia of lead poisoning is characterized by:

a) Coarse basophilic staining of RBC
b) Elevated HbA2
c) Macrocytic anemia
d) Reticulocytosis
e) Spherocytosis

Which of the following is the meaning of hypersplenism?

a) Deficient function of spleen
b) Enlarged spleen that crosses the mid-line
c) Presence of multiple spleens
d) Reduced size of spleen
e) Splenomegaly with depressed one or more blood elements

Hypersplenism is characterized by which of the following?

a) Correction of hematological abnormalities by splenectomy
b) Decreased formation of the deficient blood element in the bone marrow
c) Increase of one or more of cellular elements of the blood
d) Normal sized spleen with increased demand to blood transfusion
e) Splenomegaly with normal blood count

Features of hereditary spherocytosis include:

a) Hb electrophoresis is essential for its diagnosis
b) Microcytic hypochromic anemia
c) Negative family history
d) Onset may occur in neonatal period
e) Splenectomy increase hemolysis to a large extent

Spherocytosis is diagnosed by:

a) Elevated level of Hb F
b) Osmotic fragility test
c) Reduced level of serum iron
d) Reduced level of G6PD
e) Sickling test

Splenectomy is almost curative in cases of:

a) G6PD deficiency
b) Sickle cell anemia
c) Spherocytosis
d) Thalassemia major
e) Thalassemia trait

Which of the following conditions presents as hemolytic anemia in neonatal period?

a) G6PD deficiency
b) Sickle cell anemia
c) Thalassemia
d) TORCH infection
e) Vitamin K deficiency

Antibiotics that can precipitate hemolysis in G6PD deficiency include:

a) Aminoglycosides
b) Cephalosporin
c) Erythromycin
d) Penicillin
e) Sulfonamide

A 6-year-old boy with thalassemia major was found to have an enlarged tender liver. This could be caused by:

a) Anemic heart failure
b) Extramedullary hematopoiesis
c) Hemosiderosis
d) Hypersplenism
e) Liver cirrhosis

A 6-year-old girl with thalassemia major was found with severe jaundice. This could be due to:

a) Aplastic crisis
b) Extramedullary hematopoiesis
c) Formation of urinary stones
d) Hemosidrosis of the liver
e) Hepatitis A infection due to transfusion of infected blood

The curative treatment for patients with thalassemia major is:

a) Bone marrow transplantation
b) Colony stimulating factor
c) Hydroxy urea
d) Iron chelation therapy
e) Splenectomy

A 2-year-old boy was diagnosed as severe hemophilia A. On examination you may find:

a) Extensive purpuric rash
b) Generalized lymphadenopathy
c) Hemoarthrosis
d) Hepatosplenomegaly
e) Hypertension

Diagnosis of hemophilia A is suggested by laboratory investigation if there is:

a) Defective platelets function
b) Defective platelets number
c) Prolonged partial thromboplastin time
d) Prolonged prothrombin time
e) Prolonged thrombin time

Hemoglobin electrophoresis in sickle cell anemia shows:

a) Hb A is 80 – 95 %
b) Hb A2 is 80 – 95 %
c) Hb F is 80 – 95 %
d) Hb H is 80 – 95 %
e) Hb S is 80 – 95 %

Hyperhemolytic crisis is precipitated with:

a) Exposure to blood transfusion

b) Exposure to hepatitis B infection
c) Exposure to hypoxia
d) Exposure to oxidant agent
e) Exposure to parvovirus B19 infection

Aplastic crisis is precipitated with:

a) Exposure to blood transfusion
b) Exposure to dehydration
c) Exposure to hypoxia
d) Exposure to oxidant agent
e) Exposure to parvovirus B19 infection

Vasoocclusive crisis is precipitated with:

a) Exposure to blood transfusion
b) Exposure to drugs
c) Exposure to hypoxia
d) Exposure to oxidant agent
e) Exposure to parvovirus B19 infection

Which of the following is a cause of platelet function disorders?

a) Henoch- Schonleinpurpura
b) Leukemia
c) Scurvy
d) Thrombocytosis
e) Uremia

Drugs causing defective platelets aggregation include:

a) Acetaminophen
b) Carbamazipin
c) Chloramphenicol
d) Salicylates
e) Sulphonamide

Besides purpura, manifestations of Henoch-Schonleinpurpura may include:

a) Cough
b) Generalized lymphadenopathy
c) Hematuria
d) Hepatosplenomegaly
e) Jaundice

The following is clinical feature of anaphylactoid purpura:

a) Ascites
b) Hepatitis
c) Huge splenomegaly
d) Intermittent colicky abdominal pain
e) Pancreatitis

Henoch-Schonleinpurpurais a type of:

a) Hematologic malignancy
b) Non-thrombocytopenic purpura

c) Pancytopenia
d) Thrombasthenia
e) Thrombocytopenic purpura

Which of the following is true regarding prognosis of ITP?

a) 20% of cases pass into malignancy
b) 70-80% of cases show spontaneous recovery
c) Fatality rate is high
d) Intracranial hemorrhage is high
e) More than 50% of cases pass into chronic ITP

A newborn boy is noted to have hemoglobin F predominance on his newborn screen. He is otherwise well. A family history is not available. You will tell the family that this newborn:

a) Has thalassemia major
b) Is normal and send him home
c) Needs more investigations
d) Needs assessment of hemoglobin level now
e) Repeat the test in other lab

Which of the following disorders may present with abdominal pain in children?

a) Aplastic anemia
b) Folic acid deficiency anemia
c) Pure red cell anemia
d) Sickle cell anemia
e) Vitamin B12 deficiency anemia

Initial management lines of moderately severe reaction of blood transfusion include:

a) Continue the same blood but more slowly
b) Give epinephrine subcutaneously
c) Give furosemide intravenously
d) Give hydrocortisone intravenously
e) Startbroad spectrum antibiotics

A 10 years old girl presented with fever and bruises for the last 2 weeks. On examination she is pale with scattered purpura allover her body but no hepatosplenomegaly or lymphadenopathy. Her Hb is 6g/dl, WBC count is 2,700/mm^3 and platelets count is 90,000/mm^3. The most appropriate investigation to diagnose the case is:

a) Bleeding time
b) Bone marrow aspiration
c) Factor VIII assay
d) Hemoglobin electrophoresis
e) Prothrombin time and concentration

Which of the following laboratory results match diagnosis of iron deficiency anemia?

a) Serum ferritin is decreased, serum iron is decreased, and total iron binding capacity is decreased
b) Serum ferritin is decreased, serum iron is decreased, and total iron binding capacity is increased
c) Serum ferritin is decreased, serum iron is increased, and total iron binding capacity is increased
d) Serum ferritin is increased, serum iron is decreased, and total iron binding capacity is increased
e) Serum ferritin is increased, serum iron is increased and total iron binding capacity is increased

Tuberculous lymphadenopathy is suggested if there is:

a) Acute onset of enlarged lymph nodes
b) Enlarged and matted lymph nodes with sinus formation
c) Enlarged, hard and fixed lymph nodes
d) Enlarged, painful and tender lymph nodes
e) Gradual onset of enlarged hard lymph nodes

Which of the following is a good prognostic factor of acute lymphoblastic leukemia?

a) Age < 1 year
b) Female sex
c) Initial leukocytic count > 50 000mm3
d) No CNS infiltration
e) Presence of chromosomal abnormalities

A boy with acute lymphoblastic leukemia is assigned to be of high-risk group if he had got which of the following features?

a) Age 1 - 10 years
b) Early response to treatment
c) Initial leukocytic count <50 000mm3
d) No CNS infiltration
e) Presence of chromosomal abnormalities

Hodgkin disease is characterized by:

a) It is more common in infancy and early childhood
b) It is the most common malignancy in children
c) Lymph nodes are enlarged, matted and tender
d) Mediastinal syndrome is a common presentation
e) The main presentation is bone pain and bone marrow failure

The expected response to iron therapy is:

a) Gradual increase in Hb A2
b) Hemoglobin level returns to normal within 3 months
c) Improved appetite within one week.
d) Increased platelet count
e) Reticulocytosis starts 72-96 hours with a peak at 5-7 days.

A 3-year-old girl presents to pediatric clinic with a 3-week history of lethargy and weight loss. On examination she is pale and has widespread bruising. She has no other medical history and is not currently on any medications. Full blood count reveals: Hb 6 g/dl, WBC 43.000 cell/cm and platelet 50.000 cell /ml. Which of the following is the most probable diagnosis?

a) Acute lymphoblastic leukemia
b) Haemophilia A
c) Hepatic dysfunction
d) Idiopathic thrombocytopenic purpura
e) Vitamin K deficiency

During blood transfusion to a child, you notice flushing of face, rigors, tachycardia, fever and severe itchy rash. Which of the following is the first step of management?

a) Continue transfusion and assurance
b) Give chlorpheniramine orally
c) Give antipyretic orally
d) Slow the transfusion
e) Stop the transfusion

A 5-year-old boy presents to hospital with a recent history of easy bruising. Two weeks ago, he had an upper respiratory tract infection which resolved spontaneously. Today he is afebrile but has many, widespread bruises with some scattered petechiae. Investigations reveal: Hb 13 g/dl, WBC 10.000 cell/cm, platelet count: 32.000 cell/ml. Which of the following statements regarding the likely diagnosis is correct?

a) Hemoarthrosis commonly occurin this disorder
b) Iron chelating therapy is indicated
c) Prothrombin time and activated partial thromboplastin time are prolonged
d) Splenectomy is indicated
e) Spontaneous recovery within weeks to months is expected

A 4-year-old girl, known to have sickle cell anemia, acutely developed severe hand and foot pains, which followed an attack of severe diarrhea. Which type of crisis she is suffering from?

a) Acute hemolytic crisis
b) Aplastic crisis
c) Hyperhemolytic crisis
d) Sequestration crisis
e) Vaso-occlusive crisis

A 4-year-old boy presents with pallor and easy fatigability. Mother gives history of eating dirt. His platelets count is 670000, Hb is 6 g/dl, MCV is 57 fl and WBCs is 6000. What is the most likely diagnosis?

a) Anemia of chronic disease
b) G6PD Deficiency
c) Hereditary Spherocytosis
d) Iron Deficiency Anemia
e) Thalassemia Major

A 6-year-old boy who was previously well, presents with low grade fever, colicky abdominal pain, painful knee joints and a rash mainly on his legs and buttocks. On examination there was no abdominal tenderness or masses. What is the most likely diagnosis?

a) Anaphylactiodpurpura
b) Idiopathic thrombocytopenic purpura
c) Juvenile idiopathic arthritis
d) Leukemia
e) Scarlet fever

A 4-year-old previously well boy is brought to the ER by his mother. She reports that he developed pallor, red colored urine, and jaundice over the last 2 days. He has eaten fava beans with his family. What is the most likely diagnosis?

a) Autoimmune hemolytic Anemia
b) G6PD Deficiency
c) Hepatitis A
d) Hereditary Spherocytosis
e) Thalassemia Minor

A 6-year-old child presented with mild jaundice since birth. On examination the child has anemia associated with a moderate splenomegaly. A peripheral blood smear showed spheroid RBC which is abnormally small in size and lacked the normal central zone of pallor. Management of this case includes:

a) IV Fluids and analgesics
b) Oral iron is helpful
c) Oral steroids
d) Splenectomy can be curative
e) Sulfonamide are better avoided

A 2 -year-old boy presents in severe pain and swelling of hands and feet. He is afebrile but obviously in pain and distressed. Initial lab works show Hb 8.5 gm/dl, with normal MCV, MCH. His sister has history of recurrent pains of her legs and back. What is the MOST likely diagnosis?

a) Anaphylactoid purpura
b) Juvenile idiopathic arthritis
c) Sickle cell anemia
d) Thalassemia
e) Vitamin D deficiency

A 2-year-old child presents with painful swelling of the hands and feet. Laboratory evaluation reveals hemoglobin of 9 g/dL and reticulocyte count of 3%, white blood cell count of 11,500 and platelet count of 250,000. Which additional laboratory test will support your diagnosis?

a) Bone marrow aspiration
b) Hemoglobin electrophoresis
c) Osmotic fragility test
d) Serum ferritin measurement
e) Skeletal survey

Diagnosis of beta Thalassemia is established by:

a) HbA estimation.
b) Hb electrophoresis.
c) Osmotic fragility test
d) Serum iron and iron binding capacity
e) Sickling test

Purpura of the Henoch–Schoenlein type is characterized by

a) It begins as maculopapular rash and progress to palpable purpura
b) The purpura blanch with pressure
c) The purpura is not palpable
d) The rash predominantly affects the trunk and face
e) There is a purpuric rash with signs of systemic shock

A 7-year-old child with thalassemia is prepared for splenectomy operation. Which of the following vaccines is recommended to be given before operation?

a) Hepatitis A
b) MMR
c) Pneumococcal
d) Varicella
e) Viral influenza

Which of the following tests is most effective in diagnosing hemophilia A?

a) Bleeding time
b) Complete blood count (CBC)
c) Partial thromboplastin time (PTT)
d) Platelet count
e) Prothrombin time and concentration

A male patient is presented with pallor, bleeding and recurrent fever. This patient is considered to have which of the following?

a) Hemolytic anemia
b) Leukocytosis
c) Pancytopenia
d) Thrombocytopenia
e) Von Willebrand disease

Which of the following is an example of macrocytic anemia?

a) Anemia of chronic disease
b) Folic acid deficiency
c) Glucose 6-phosphate dehydrogenase deficiency
d) Iron deficiency anemia
e) Thalassemia minor

The earliest response to oral iron, in iron deficiency anemia is:

a) Changes in blood film
b) Improved neurologic function
c) Increased hemoglobin

d) Reduction in iron binding capacity
e) Reticulocytosis

A feature that signifies diagnosis of β-thalassemia minor over the diagnosis of iron deficiency anemia is:

a) Elevated hemoglobin A2
b) Increased iron binding capacity
c) Increased red cell distribution width
d) Low serum iron level
e) Normal hemoglobin F

Infection with which of the following viruses can cause aplastic crises in chronic haemolytic disease?

a) Cytomegalovirus
b) Herpes virus
c) Parvovirus B19
d) Rota virus
e) T-lymphocyte virus

The best treatment option for sever hereditary spherocytosis is:

a) Corticosteroids
b) Cyclosporine
c) Hydroxyurea
d) Immunoglobulin
e) Splenectomy

The MOST common site of relapse of acute lymphoblastic leukemia is:

a) Bone marrow
b) Brain
c) Kidneys
d) Lungs
e) Testes

Nephrology:

Properties of normal urine include

a) Freshly voided urine is white in colour
b) It contains no visible deposits
c) It is cloudy in its aspect
d) Specific gravity of urine collected over 24 hour is 1000 – 1005
e) Volume of urine is almost fixed in all age groups

The average insensible fluid loss per day is:

a) 200 ml/m2
b) 300 ml/kg
c) 400 ml/kg
d) 400 ml/m2

e) 600 ml/m2

Clinical manifestations of lower urinary tract infection include:

a) Dysuria and frequency
b) Hypertension
c) Lion pain and tenderness
d) Renal failure
e) Vomiting and diarrhea

Characters favoring glomerular origin of hematuria include:

a) Normal RBCs morphology
b) Proteinuria < 100 mg/dl
c) RBCs casts
d) Red urine with blood clots
e) Tubular epithelial casts

A 4-year-old boy is brought to the clinic because of dysuria, frequency and incontinence. For diagnosis of urinary tract infection, the following is true:

a) Culture sensitivity is done before starting therapy.
b) Culture shows 10,000 colonies in asymptomatic child is diagnostic
c) Culture shows 100,000 colonies of different organisms is diagnostic
d) Pyuria is diagnostic of urinary tract infection
e) The presence of proteinuria exclude the diagnosis

Which of the following is TRUE about clinical features of minimal change nephrotic syndrome?

a) Girls are more frequently affected than boys
b) Hematuria, hypertension and azotemia are common
c) Onset is sudden
d) Pleural effusion never occur
e) The usual age is 2-6 years

Which of the following is correct as regard prognosis of idiopathic nephrotic syndrome?

a) Children with focal sclerosis type show the best prognosis
b) Children with steroid resistance never progress to end stage renal failure
c) Less than 70% of minimal change NS patients respond to steroid
d) Most children with steroid responsive nephrotic syndrome have repeated relapses
e) The relapses are expected to increase in frequency as the child grows older.

Which of the following is a feature of minimal change nephrotic syndrome?

a) Generalized edema is mainly due to loss of albumin in urine
b) It is present in 65% of cases of idiopathic nephrotic syndrome
c) Poor response to steroids
d) There is heavy non selective proteinuria
e) There is hypocholesterolemia

A previously healthy 5-year-old girl has acute onset of edema and oliguria. Laboratory studies reveal hypoalbuminemia and hypercholesterolemia. Which of the following is the major complication with this condition?

a) Acute renal failure
b) Bacterial peritonitis
c) CNS infection
d) Congestive heart failure
e) Hypertension

A 9- year-old boy is complaining of frequent recurrent attacks of generalized edema in spite of compliance to treatment. He is hypertensive. Lab investigations revealed that serum albumin is 1.5 gm/dl, urinary protein is 70 mg/m2/hour, and serum cholesterol is 350 mg/dl.Which of the following is the recommended next step for diagnosis?

a) Abdominal ultrasound
b) Renal biopsy
c) Renal scanning with technetium
d) Total plasma proteins
e) Urine creatinine

A Five-year-old boy presented with gradual onset of puffy eyes and edema of lower limbs. Urine protein is 60 mg/m^2/hour (normal figure is 4 mg/ m^2/hour) and serum cholesterol is 300 mg/dl (normal figure is 120-200 mg/ dl). Before you start treatment, it is important to give the patient the following vaccine:

a) DPT vaccine
b) Hemophilus Influenza type B vaccine
c) Hepatitis A vaccine
d) Intramuscular polio vaccine
e) Pneumococcal vaccine

A 4-year-old boy presents to the hospital with a chief complaint of swelling around the eyes and ankles. You note the patient to have pitting edema up to the knees bilaterally and peri-orbital edema. Urineanalysis shows 3+ protein in the urine. What other laboratory findings in this case you may find?

a) Hypercholesterolemia
b) Hyperglobulinemia
c) Hypocomplementemia C3
d) Pus cells in urine > 5cells/high power field
e) Raised erythrocyte sedimentation rate

Clinical features of minimal change nephrotic syndrome include:

a) Acute onset of hematuria and oliguria.
b) Edema initially noted around eyes and lower limbs.
c) Girls are affected more than boys.
d) Most commonly it appears since birth.
e) Non-pitting generalized edema.

A 4 -year-old patient was brought to the clinic because of generalized swelling of lower limbs without other symptoms. According to the mother there was a similar attack since 1year. The patient was receiving prednisolone on an alternate day regimen that was stopped since 4 months. On exam: BP 90/60 mmHg, pitting edema of lower limbs. Urine with dipstick: protein 4+, no other abnormalities. This case is considered as which of the following?

a) Frequent relapse nephrotic syndrome
b) Infrequent relapse nephrotic syndrome
c) Secondary nephrotic syndrome
d) Steroid dependent nephrotic syndrome
e) Steroid resistant nephrotic syndrome

A 9-year-old has hematuria and an increased serologic titer to antistreptolysin O (ASO). What will be the most likely finding?

a) Decreased serum C3
b) Decreased urinary protein/creatinine ratio
c) Heavy proteinuria
d) IgA deposits in kidney biopsy
e) Increased serum cholesterol

Which of the following causes red urine without RBCs?

a) Heavy exercise
b) Hemolytic uremic syndrome
c) Pyelonephritis
d) Trauma
e) Urates

The first line drug to be prescribed in treatment of acute post streptococcal gromerulonephritis is:

a) Corticosteroids
b) Furosemide
c) Penicillin
d) Rifampicin
e) Trimethoprim – sulfamethoxasole

Heavy proteinuria is defined as

a) Spot urine protein to creatinine ratio less than 2
b) The presence of hyaline cast in urine sample
c) The presence of polymorphonuclear leucocytes in urine sample
d) Urinary protein excretion exceeds 20 mg/m2/hour
e) Urine protein > 3.5 gm/24 hour

In acute renal failure, hyperkalemia is treated by:

a) Intravenous calcium gluconate
b) Intravenous immunoglobulin
c) Intravenous saline solution
d) Intravenous salt free albumin
e) Intravenous sodium bicarbonate

Endocrinal disorders

Which of the following roentgenographic findings in a newborn infant is most suggestive of undiagnosed hypothyroidism?

a) Absent distal femoral epiphysis at birth
b) Absent proximal femoral epiphysis at birth
c) Absent ossification center of head of humerus at birth
d) Closure of cranial sutures at birth
e) Two carpal bones at birth

Congenital hypothyroidism should be included in the differential diagnosis of a newborn with which of the following?

a) Anemia
b) Elevated liver enzymes
c) Leukocytosis
d) Prolonged jaundice
e) Thrombocytopenia

Diabetic ketoacidosis is diagnosed when blood gas analysis shows which of the following?

a) pH 7.1 & bicarbonate 11 mmol/l
b) pH 7.1 & bicarbonate 33 mmol/l
c) pH 7.35 & bicarbonate 22 mmol/l
d) pH 7.5 & bicarbonate 11 mmol/l
e) pH 7.5 & bicarbonate 33 mmol/l

A 4-year-old girl is admitted with history of fever, vomiting and abdominal pain for 2 days. Clinically, the child is sleepy, dehydrated and acidotic breathing, with a heart rate of 122/min and blood pressure of 90/ 48 mmHg. Her blood sugar is 480 mg%. Which of the following is the expected pattern of respiration?

a) Alternating shallow and deep
b) Rapid, deep
c) Rapid, shallow
d) Slow, deep
e) Slow, shallow

A 14-year-old adolescent girl was monitored for 7 years for a history of Type-1 diabetes mellitus. Glycosylated hemoglobin was done revealing high result. This laboratory test indicates which of the following?

a) Her glucose control is poor
b) She does not have insulin-dependent diabetes
c) She has an underlying infection
d) She has diabetic ketoacidosis
e) Within normal blood glucose level

Diabetic ketoacidosis is one of the differential diagnosis if the child is presented with:

a) Diarrhea
b) hypertension

c) Oliguria and proteinuria
d) Shallow and slow respiration
e) Vomiting and abdominal pain

Diabetic ketoacidosis is one of the differential diagnosis if the child is presented with:

a) Cheyne stokes respiration
b) Prolonged expiration
c) Rapid deep respiration
d) Respiratory distress
e) Shallow and slow respiration

Biochemical diagnosis of Type 1Diabetes Mellitus includes:

a) Fasting blood sugar ≥ 126 mg/dl.
b) PH level is less than 7.3
c) Random plasma glucose < 300 mg/dl
d) Serum bicarbonate level < 15 mmol/ L
e) Two hours plasma glucose in oral glucose tolerance test > 200 mg/dl

The disturbed level of consciousness in diabetic ketoacidosis is caused by:

a) Acidosis
b) Hyperglycemia
c) Hyperkalemia
d) Hypernatremia
e) Increase serum level of total lipids

A 5-year-old male presents with a 5-lb weight loss over the previous 2 weeks and nighttime enuresis. He has vomiting since yesterday. On examination, he is alert and talkative with a BP of 90/60 mm Hg and a heart rate of 130 beats per minute. Mucous membranes are sticky, and respirations are rapid and deep.The most likely diagnosis is:

a) Acute gastroenteritis complicated by dehydration
b) Acute lobar pneumonia
c) Diabetic ketoacidosis
d) Drug intoxication
e) Urinary tract infection

A 5-year-old male presents with a 5-lb weight loss over the previous 2 weeks and nighttime enuresis. On examination, he is alert and talkative with a BP of 90/60 mm Hg and a heart rate of 130 beats per minute. Mucous membranes are sticky, and respirations are rapid and deep. Laboratory finding to confirm diagnosis includes:

a) Blood PH between 7.35 – 7.45
b) Blood glucose level < 200 mg/dl
c) Fasting blood glucose level < 126 mg/dl
d) Serum bicarbonates less than 15mmol/l
e) Serum potassium level > 5.6 mmol/dl

While educating the mother of a 4-year-old child with type 1 diabetes, you advise her about the meal plans that:

a) All sucrose and sugars should be stopped
b) Carbohydrates should not exceed 30% of the meal
c) 70% of carbohydrate should be of the complex form
d) Fat should be decreased to10% of the meal
e) Proteins should be increased to 50% of the meal

While educating the mother of a 10-year-old girl with type 1 diabetes, you advise her about the meal plans that:

a) Animal fat is better than plant fat
b) High fiber diet is useful for controlling blood glucose
c) Meals should be restricted to the 3 main meals only
d) Sugary diet is better than complex carbohydrates diet
e) The main constituent of the meal should be protein and fibers

A 12-year-old boy with type 1 diabetes is aiming to join the school football team. About exercises you advised him to:

a) Avoid all exercises for fear of hypoglycemia
b) Avoid strenuous exercise if there is ketonuria
c) Increase the dose of insulin by 10-15% in the day of exercise
d) No precautions should be taken with exercise
e) Stop insulin in the day of exercise

Laboratory monitoring of type 1 diabetes includes:

a) Glycosylated hemoglobin every year
b) Lipid profile after 5 years of diagnosis
c) Liver function test every year
d) Microalbumenuria after 1 year of diagnosis
e) TSH and fT4 every year

A 3-week-old baby is referred to you because of prolonged neonatal jaundice, somnolence and shocking spells. On examination you noticed jaundice, noisy respiration and open posterior fontanel. What is the expected diagnosis?

a) Congenital hypothyroidism
b) Crigler-Najjar syndrome
c) Infant of diabetic mother
d) Kernicterus
e) Spherocytosis

A 3-week-old baby is referred to you because of prolonged neonatal jaundice, somnolence and shocking spells. On examination you noticed jaundice, noisy respiration and open posterior fontanel. You ordered serum TSH and fT4 levels. The expected results are:

a) High TSH and high T4 levels
b) High TSH and low T4 levels
c) Low TSH and high T4 levels
d) Low TSH and low T4 levels
e) Normal TSH and T4 levels

A 3-week-old baby is referred to you because of prolonged neonatal jaundice, somnolence and shocking spells. On examination you noticed jaundice, noisy respiration and open posterior fontanel. Investigations showed high serum TSH and low fT4 levels. The next step is to:

a) Assurance of the mother
b) Ordered phototherapy
c) Prescription of L-thyroxin
d) Reassessment of investigations after 1 month
e) Stop breast feeding

Screening newborn for congenital hypothyroidism is characterized by:

a) High TSH screening level indicate immediate treatment with L-thyroxin
b) High TSH screening level is considered suspect and should confirmed
c) In Egypt it is done by assessment of fT4 level
d) In confirmed hypothyroid cases, treatment starts after the 1st month of life
e) It is done on the first day of life

Familial short stature is characterized by:

a) Birth length is short
b) Positive family history of short stature
c) Ultimate normal adult height
d) Growth hormone is deficient
e) Delayed puberty

Constitutional short stature is characterized by:

a) Growth hormone is deficient
b) Positive family history of short adult stature
c) Puberty is started at the proper time
d) Small for gestational age at birth
e) Ultimate normal adult height

Treatment of congenital hypothyroidism with L-thyroxin is characterized by:

a) If the patient has iron deficiency anemia, mix thyroxin with iron therapy
b) It is given by subcutaneous injection
c) Monitoring of the dose is by hemoglobin level every 3 months
d) The dose per Kg is increased as the patient gets older
e) Treatment is for life

Genetics

Chromosomal disorders should be suspected in patients with:

a) Consanguineous marriage
b) Multiple congenital anomalies
c) Multiple pregnancy
d) Old mother
e) Preterm birth

As regard to chromosomal non disjunction:

a) An example of monosomy is turner syndrome
b) It almost occur at the level of gametes formation
c) The most common monosomy is down syndrome
d) When occurs in meiotic division results in mosaicism
e) When occurs in mitotic division results in monosomy and trisomy

Which of the following is a character of autosomal recessive inheritance?

a) Abnormalities of chromosomal number
b) Consanguinity has no role
c) Every affected person has an affected parent
d) Males and females are equally affected
e) No male to male transmission

The mode of inheritance in cases of hemophilia A is:

a) Autosomal dominance
b) Autosomal recessive
c) Chromosomal abnormalities
d) X linked dominant
e) X linked recessive

In X linked recessive inheritance:

a) Females who carry one gene are diseased
b) Males who carry one gene never express the disease
c) Spherocytosis is an example
d) The carrier mother transmits the disease to her sons
e) The diseased father transmits the disease to his sons

Occurrence of Down syndrome is characterized by:

a) Girls are affected more than boys
b) Mosaicism increases with aging of the ovum
c) Translocation is the most common type
d) The risk of Down syndrome in general population is 1/700
e) Translocation Down syndrome increases with advancing maternal age

Which of the following is true regarding the genetic types of Down syndrome?

a) Clinically translocation Down syndrome differ from non-disjunction type
b) In the non-disjunction, the fertilized ovum has 23 chromosomes
c) Mosaicism Down syndrome has a single cell line
d) The number of chromosomes in regular type is 46
e) The translocation type has 3 copies of chromosome 21

A 4-month-old female infant has generalized hypotonia, small ears, inner epicanthal folds, clinodactyly, and wide-spaced first and second toes. For which of the following problems is she most at risk?

a) Chorioretinitis
b) Hearing loss
c) Hydrocephalus

d) Leukemia
e) Seizure disorder

A 40 - years- old lady gave birth to her 7th child who was low birth weight, he had epicanthic folds, malformed low set ears and up slanted palpebral fissure. Which of the following is the most likely mechanism of this abnormality?

a) Balanced translocation
b) Mosaic type
c) Non-disjunction
d) Autosomal recessive inheritance
e) Autosomal dominant inheritance

A 2-days- old baby was born to a 40- year-old mother. On examination this baby has low set ears, bilateral simian crease, clinodactyly, slanted palpebral fissures, and a heart murmur.Which of the other signs would be expected in this baby?

a) Coarse hair
b) Dry and thick skin
c) Hypotonia
d) Small tongue
e) Webbing of the neck

Complications of Down syndrome include:

a) Anemia
b) Liver cirrhosis
c) Nephrotic syndrome
d) Recurrent respiratory infections
e) Renal failure

Skull examination of a case of Down syndrome shows:

a) Brachycephaly
b) Hydrocephalus
c) Macrocephaly
d) Oxycephaly
e) Scaphocephaly

The risk of recurrence of Down syndrome after having a baby with trisomy 21 (non-dysjunction) is:

a) 1/3-5
b) 1/700
c) 100/100
d) 25/100
e) 50/100

The risk of recurrence of Down syndrome after having a baby with inherited translocation type is:

a) 1/3-5
b) 1/700
c) 25/100
d) 50/100
e) 100/100

Which of the following is the mode of inheritance of spherocytosis?

a) Autosomal dominant
b) Autosomal recessive
c) Translocation
d) X linked dominant
e) X linked recessive

Which of the following is the mode of inheritance of Thalassemia?

a) Autosomal dominant
b) Autosomal recessive
c) Translocation
d) X linked dominant
e) X linked recessive

Which of the following is the mode of inheritance of G6PD?

a) Autosomal dominant
b) Autosomal recessive
c) Translocation
d) X linked dominant
e) X linked recessive

Nervous system

A 9-month-old infant was diagnosed as cerebral palsy as a complication of kernicterus during neonatal period. Which of the following is the expected type of cerebral palsy?

a) Ataxic
b) Extrapyramidal
c) Spastic diplegia
d) Spastic hemiplegia
e) Spastic quadriplegia

Which of the following is the best anticonvulsant drug used in neonatal seizures?

a) Carbamazepine
b) Diazepam
c) Phenobarbital
d) Phenytoin
e) Valproate

Which of the following is an etiology of metabolic causes of acute seizures?

a) Hypercalcemia
b) hyperchloremia
c) Hyperglycemia
d) Hypermagnesemia
e) Hypernatremia

An otherwise normal 2-year-old boy has a brief, self-limited seizure associated with an elevated temperature. Hisexamination showed follicular tonsillitis. He has completely recovered within 1 to 2 hours of the seizure.Which of the following is a risk factor for a child to develop epilepsy?

a) Abnormal EEG
b) Convulsion lasting for 5 minutes
c) Focal convulsions
d) Generalized convulsions
e) Positive family history of febrile convulsions

Risk factor to develop epilepsy as a complication of febrile convulsions is:

a) Abnormal EEG
b) Initial febrile seizure after 1 year of age
c) Patients not receiving preventive anticonvulsants
d) Positive family history of epilepsy
e) Typical febrile seizure

Which of the following is a characteristic feature in spastic diplegic cerebral palsy?

a) Circumduction gait is characteristic
b) High association of mental retardation and seizures
c) It affect one side of the body
d) Lower limbs are more affected than the upper limbs
e) Upper limbs are more affected than the lower limbs

Which of the following is a clinical feature of extrapyramidal cerebral palsy?

a) Athetoid movements is noticed since neonatal period
b) Infants are characteristically hypertonic
c) Intellect is always affected
d) It can complicate cases of kernicterus
e) May follow hyperglycemia

Cerebral palsy is characterized by:

a) It involves posture and movements
b) It is a progressive disorder of cerebral function
c) It is caused by a genetic defect
d) It is often associated with normal speech and vision
e) It results from insults to the brain in late childhood

Which of the following is a treatable cause of mental retardation?

a) Cerebral palsy
b) Congenital rubella

c) Down syndrome
d) Hypothyroidism
e) Mucopolysaccharidosis

Which of the following is a screening test for detection of preventable causes of mental retardation?

a) Chromosomal study
b) CT or MRI brain
c) Placental function
d) Reducing substance in the urine
e) Serologic test for rubella infection

A pregnant mother who has 2-year-old mentally retarded child asks about neonatal screening tests for prevention of mental retardation in the following baby. Of the following, what screening test is indicated?

a) CBC for iron deficiency anemia
b) EEG for epileptic activity
c) Karyotyping
d) Neonatal TSH assessment
e) Skull X ray for intracranial calcification

Which of the following is an etiology of increased intracranial pressure?

a- Cerebral palsy
b- Febrile convulsion
c- Hypovitaminosis A
d- Intracranial hemorrhage
e- Microcephaly

A 2-day-newborn was delivered by normal vaginal delivery. Head circumference is 28 cm. Which of the following is a feature of this head size?

a) Beaten silver appearance is a characteristic X ray finding
b) Characterized by normal intelligence
c) Delayed closure of fontanels
d) Serial skull measurementsshow accelerated growth
e) Seizures and spasticity may be associated

A 2-day-newborn was delivered by normal vaginal delivery. Head circumference is 30 cm. Which of the following is a possible etiology?

a) Cephalohematoma
b) Chronic hemolytic anemia
c) Down syndrome
d) Hydrocephalus
e) Rickets

Which of the following is a cause of secondary microcephaly?

a) Craniosynostosis
b) Down syndromes
c) Familial
d) Hydrocephalus

e) Neonatal asphyxia

A 4-year-old child comes to your clinic for the first time walking withthe help of leg braces and a walker, and his speech is slurred and limited to short phrases. He hasnever been hospitalized. On examination youfind that the boy has only minimally increased tone in the upper extremities but good fine motorcoordination; hehas significantly increased tone and deep tendon reflexes in the lower extremities. Which of the following types of cerebral palsy is the diagnosis of this case?

a) Ataxic type
b) Extrapyramidal type
c) Spastic diplegia
d) Spastic hemiplegia
e) Spastic quadriplegia

A 4-year-old child comes to your clinic for the first time walking withthe help of leg braces and a walker, and his speech is slurred and limited to short phrases. He hasnever been hospitalized. On examination youfind that the boy has only minimally increased tone in the upper extremities but good fine motorcoordination; hehas significantly increased tone and deep tendon reflexes in the lower extremities. Which of the following is a character of this disease?

a) It is a hereditary disorder
b) May follow perinatal insult
c) Never associated with vision affection
d) No risk of epilepsy
e) Slowly progressive motor deficit

A 4-year-old child comes to your clinic for the first time walking with the help of leg braces and a walker, and his speech is slurred and limited to short phrases. He has never been hospitalized. On examination you find that the boy has only minimally increased tone in the upper extremities but good fine motor coordination; he has significantly increased tone and deep tendon reflexes in the lower extremities. Which of the following isan etiology of this disease?

a) Congenital cytomegalovirus infection
b) Hyperglycemia
c) Hypothyroi
d) Thalassemia major
e) Iron deficiency anemia

An otherwise normal 2-year-old boy has a brief, self-limited seizure associated with an elevated temperature. His examination showed follicular tonsillitis. He has completely recovered within 1 to 2 hours of the seizure. Which of the following support the diagnosis of febrile convulsions?

a) History of neonatal convulsion
b) It occurs at the onset of rapid rise of temperature
c) Most cases have positive family history of febrile convulsions
d) Simple febrile convulsion lasts > 15 minutes
e) Usually associated with signs of meningeal irritation

Collagen diseases:

Criteria of diagnosis of juvenile rheumatoid arthritis include:

a) Acute onset of arthritis
b) Age of onset > 16 years
c) Duration of arthritis > 6 weeks
d) Involvement of < 5 joints is an exclusion
e) Involvement of hip joint is diagnostic

Systemic corticosteroids are used for treatment of JRA in which of the following conditions?

a) Polyarticular type
b) Raised ANA titer
c) The presence of hepatosplenomegaly
d) The presence of pericarditis
e) The presence of rheumatoid nodules

Which type of juvenile rheumatoid arthritis is mostly associated with rheumatoid nodules:

a) Pauciarticular JRA, rheumatoid factor positive
b) Polyarticular JRA, ANA positive
c) Polyarticular JRA, rheumatoid factor negative
d) Polyarticular JRA, rheumatoid factor positive
e) Systemic onset JRA, ANA positive

Rheumatoid nodules are characterized by:

a) Usually associated with less than 5 joints affection
b) Usually associated with mild course
c) Usually associated with rheumatoid factor positive
d) Usually associated with rheumatoid rash
e) Usually found on flexor surface of elbow

A5- year-old girl diagnosed with pauciarticular juvenile idiopathic arthritis has a positive antinuclear antibody test. Which of the following would this patient is at increased risk to develop?

a) Encephalitis
b) Hepatitis
c) Iridocyclitis
d) Nephritis
e) Peritonitis

A 4-year-old white girl has had joint swelling in multiple joints for over 6 months. She is slow to move in the morning and moves as if stiff for the 1st hours of the day. Thereafter, she is a very active child. She has no rash and very little limitation of range of motion. The most likely diagnosis is:

a) Acute rheumatic arthritis

b) Bacterial arthritis
c) Juvenile rheumatoid arthritis
d) Henoch-Schonleinpurpura
e) Vitamin D deficiency

Growing pain of lower limbs is characterized by:

a) It is associated with restricted mobility
b) It is common in the ages 3 – 15 years
c) It is due to minute fractures of bone
d) Pain is more during daytime
e) Pain is usually localized

Growing pains of the lower limbs are characterized by:

a) Common in the first 2 years of life
b) Occur in periods of slow rate of growth
c) Pain is associated with limited joint movement
d) Pain is localized to ankle joint
e) Pain occurs during night

Which of the following juvenile rheumatoid arthritis patients carry the poorest prognosis?

a) More than 5 joints are affected
b) Involvement of hip joint
c) Involvement of lumbosacral spine
d) Polyarticular rheumatoid factor negative
e) Systemic onset type

Pediatric emergencies

During management of a shocked child, criteria for satisfactory volume resuscitation include:

a) Cool mottled extremities
b) Decreased heart rate to normal
c) Delayed capillary refill
d) Hyperventilation
e) Urine output less than 1ml/kg/hour

Compensated stage of shock in children is characterized by:

a) Decreased gut motility
b) Decreased heart rate
c) Decreased peripheral pulse
d) Decreased respiratory rate
e) Decreased urine volume

A 4-month-old infant presents with 3 days of profuse watery diarrhea and frequent vomiting. Vital signs are temp. 38C, HR 170/ minute, and BP 90/60 mm

Hg. He is oliguric with sunken eyes and fontanel, and poor skin turgor. This description is most consistent with which of the following?

a) Dehydration with normal cardiovascular state
b) Hypovolemic shock, compensated
c) Hypovolemic shock, decompensated
d) Irreversible shock
e) Septic shock, decompensated

Intravascular volume expansion in management of shock could be achieved by:

a) H2 receptor antagonists
b) IV dopamine therapy
c) IV fluid therapy
d) IV furosemide therapy
e) Oxygen therapy

Principles of management of poisoning include:

a) Activated charcoal is indicated in almost all drug toxicity
b) Gastric lavage is beneficial in poisoning with petroleum derivatives
c) Gastric lavage is indicated in cases with strong acids poisoning
d) Use of salt as an emetic
e) Petroleum derivatives toxicity is treated by forced diuresis

Gastric lavage in management of poisoningis characterized by:

a) Gastric lavage is beneficial in poisoning with petroleum derivatives
b) Gastric lavage is carried while the patient is sitting
c) Gastric lavage is done by ipecac syrup
d) Gastric lavage is effective if done within 1 hour post ingestion
e) Gastric lavage is indicated in cases with strong acids poisoning

In case of management of poisoning, gastric lavage is indicated in:

a) Comatosed patient
b) Poisoning with organophosphorus compounds
c) Poisoning with petroleum derivatives
d) Poisoning with strong acids
e) Poisoning with strong alkalies

Enhancing excretion of a poison can be achieved by:

a) Activated charcoal
b) Cathartics
c) Emesis
d) Gastric lavage
e) Hemodialysis

A 3-year-old boy presented to the ER 30 minutes after ingestion of corrosive compound. The next step you order is to:

a) Do forced diuresis
b) Do gastric lavage
c) Give activated charcoal

d) Give milk or water
e) Induce emesis

Poisoning with kerosene carry the risk of causing which of the following?

a) Aspiration
b) Diarrhea
c) Esophageal perforation
d) Myocarditis
e) Pinpoint pupil

Which of the following is given for treatment of a child with organophosphorus poisoning?

a) Acetylcysteine
b) Atropine
c) Deferoxamine
d) Milk
e) Sodium bicarbonate

Behavioral pediatrics

Which of the following is a correct option in treatment of nocturnal enuresis?

a) Avoid fluid restriction at night
b) Conditioning therapy is curative for older children
c) Holding urine for longer period during the day
d) Pharmacotherapy starts before age of 6 years
e) Punishment may give beneficial results in some cases

A 7-year-old girl develops secondary nocturnal enuresis. What is the next best study?

a) Creatinine clearance
b) Renal ultrasound
c) Urinary protein excretion
d) Urine analysis
e) Voiding cystourethrogram

Nocturnal enuresis is defined as occurrence of involuntary voiding at night after the age of

a) 3 years
b) 4 years
c) 5 years
d) 6 years
e) 7 years

A mother of 2-year-old boy is complaining that her baby has recurrent attacks of stoppage of respiration and cyanosis after an initial cry. These attacks are

precipitated by:

a) Calcium deficiency
b) Iron deficiency
c) Vitamin A deficiency
d) Vitamin B6 deficiency
e) Zinc deficiency

Breath holding spell is characterized by:

a) Common before 1 year of age
b) It starts with generalized seizure
c) It manifests by irregular breathing
d) It manifests by stoppage of breathing
e) Its clinical manifestations is due to associated congenital heart disease

Treatment of breath holding spells include:

a) Beta blockers
b) Oral calcium therapy
c) Oral iron therapy
d) Short duration of anticonvulsant
e) Vitamin D therapy

Which of the following is a line of treatment of breath holding spells?

a) Assurance of parents that it is a self limited condition
b) Putting the child on his back and start mouth to mouth breathing
c) Rectal diazepam during the attack
d) Slapping of the back after stoppage of respiration
e) Water submersion during the attack

Toilet training of children is characterized by:

a) Boys achieve bladder control earlier than girls
b) Night control is achieved earlier than day time
c) The average time for training success is 1 month
d) The proper age of toilet training is 2.5 – 4 years
e) Urinary control is achieved before bowel control

A 10-year-old male presents to your office with his parents with the chief complaint of wetting the bed. He has never had a period longer than 2 weeks in his life when he has not wet the bed. He has tried decreasing nighttime fluids and midnight awakening and this has not made a difference. His parents are looking for a solution. What therapy has been shown to have the best success rate?

a)Assurance of parents that the condition is self limited
b) Conditioning therapy (alarm/arousal system)
c) Daytime holding of urine
d) Desmopressin acetate for 1 month duration
e) Trial of some punishment

Concerning nocturnal enuresis:

a) Family history is positive in 10% of cases
b) It affects girls more than boys

c) It ceases spontaneously in most cases
d) It is the occurrence of involuntary voiding at night after 3 years of age
e) Secondary type is defined as wetting the bed after being continent for 3 months